

# Foundation Models and Reasoning LLMs for Scientific Multimodal Time Series: A Survey

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**Abstract**—Time series and spatio-temporal dynamics arising in natural systems—such as global weather, long-term climate, hydrological flood regimes, ocean circulation, earthquake wave propagation, satellite earth observations, and heliophysical radiation—differ fundamentally from generic tabular or urban traffic series. They are governed by continuous non-linear partial differential equations, multi-scale turbulent interactions, heterogeneous spatial geometries, and strict physical conservation laws. Recently, the emergence of pre-trained foundation models (PFMs) and scientific reasoning Large Language Models (LLMs) has instigated a paradigm shift across computational earth and physical sciences. This survey provides the first comprehensive, systematic review of foundation models and reasoning LLMs tailored for scientific multimodal time series. Guided by the PRISMA 2020 methodology, we review landmark architectures developed between 2021 and 2026 across major natural science domains. We introduce a systematic taxonomy structured along four orthogonal axes: (1) scientific domains and multimodal data representations, (2) foundation model architectural backbones (including 3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, and generative diffusion ensembles), (3) levels of physical law integration (from purely data-driven to soft physical penalties and hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (as direct predictors, contextual enhancers, and autonomous tool-orchestrating agents). Furthermore, we curate standardized operational evaluation protocols (such as WeatherBench 2 and SciTS), contrast model scale against forecast horizon, and identify crucial open frontiers spanning universal cross-physics tokenization, physical conservation guarantees, and trustworthy autonomous scientific discovery.

**Index Terms**—Scientific Foundation Models, Multimodal Time Series, Spatio-Temporal Modeling, Large Language Models, Scientific Reasoning, Weather and Climate Forecasting, Physics-Informed Machine Learning.



## 1 Introduction

Temporal and spatio-temporal dynamics constitute the empirical foundation of the natural sciences. From the planetary atmospheric circulation driving global weather patterns to the sub-surface acoustic vibrations heralding seismic rupture, physical processes unfold across continuous temporal trajectories and multi-scale spatial coordinates. Over the past several decades, the modeling and prediction of these scientific phenomena have predominantly relied on Numerical Weather Prediction (NWP) and classical numerical differential equation solvers, such as the Integrated Forecasting System (IFS) maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF) and finite-difference hydrodynamic models. While physically principled, numerical solvers incur colossal computational burdens, requiring massive supercomputing clusters to perform high-resolution discretization and step-by-step numerical integration.

In parallel, the broader machine learning community has witnessed a revolution driven by pre-trained foundation models (PFMs) and Large Language Models (LLMs). While foundational models initially transformed natural language processing and computer vision, their underlying principles—self-supervised pretraining on planetary-scale corpora, emergent scaling behaviors, and zero-shot downstream transferability—have rapidly migrated into temporal data mining. Earlier

general surveys, notably the seminal work by Jin et al. [1] and its position sequel [2], comprehensively charted large models for general time series analysis (LM4TS) and spatio-temporal data mining (LM4STD). However, as highlighted in Table 1, existing overviews focus heavily on urban computing, human mobility trajectories, traffic sensor networks, and generic 1D benchmark datasets (e.g., electricity, weather station benchmarks, exchange rates).

Natural physical systems diverge fundamentally from human-generated temporal sequences across several intrinsic dimensions:

- 1) **Governing Physical Laws and Conservation Constraints:** Physical time series are bounded by fundamental conservation laws of mass, energy, momentum, and moisture flux. Models that learn purely statistical correlations often generate unphysical states, such as negative precipitation or violation of energy conservation over extended rollouts.
- 2) **High-Dimensional Multimodal Representations:** Scientific investigations integrate heterogeneous observation modalities, including 3D atmospheric pressure tensors (e.g., ERA5 reanalysis with geopotential height, temperature, and wind fields), multi-spectral satellite imagery (e.g., Sentinel-2, Landsat), river gauge discharge records, and continuous seismic waveforms.
- 3) **Continuous Spatio-Temporal Geometries:** The Earth is a non-Euclidean spherical geometry with topological singularities at the poles, rendering standard 2D

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TABLE 1  
Comparison of This Survey with Existing Literature and Related Temporal Surveys.

| Survey / Study  | Year | Scope Focus             | Physical Grounding | Multimodal Modalities                  | LLM Reasoning                  | Core Domains Covered                                                            |
|-----------------|------|-------------------------|--------------------|----------------------------------------|--------------------------------|---------------------------------------------------------------------------------|
| Jin et al. [1]  | 2023 | General Spatio-Temporal | Moderate           | General Multimodal                     | Limited                        | Urban traffic, human trajectory, generic time series                            |
| Jin et al. [2]  | 2024 | LLM for Time Series     | Low                | Text + 1D Series                       | Moderate (Position)            | Benchmark time series forecasting, tokenization                                 |
| Rasp et al. [3] | 2024 | Weather Benchmark       | High (Meteorology) | Gridded NWP Reanalysis                 | None                           | Global numerical and ML weather forecasting                                     |
| This Survey     | 2026 | Scientific Multimodal   | High (PDEs & Laws) | Gridded + Satellite + Waveforms + Text | Comprehensive (Agents & SciTS) | Weather/Climate, Earth Observation, Hydrology, Oceans, Seismology, Heliophysics |

Euclidean convolutions and planar patch projections geometrically distorted.

- 4) The Need for Scientific Reasoning and Autonomous Discovery: Beyond numerical point prediction, domain scientists demand causal attribution, physical hypothesis testing, and interactive tool orchestration. This has catalyzed the emergence of scientific reasoning LLMs and autonomous scientific agents capable of operating over complex temporal data.

To bridge this literature gap, this survey delivers the first systematic, domain-science-centered investigation of foundation models and reasoning LLMs developed for scientific multimodal time series. We synthesize breakthroughs from 2021 through 2026 across atmospheric sciences, Earth observation, hydrology, oceanography, seismology, and heliophysics.

### 1.1 Contributions of This Survey

The primary contributions of this survey are summarized as follows:

- Domain-Science Centered Scoping: We delimit the survey strictly to natural physical systems, explicitly analyzing the coupling between continuous physics, governing equations, and deep representation architectures.
- Four-Dimensional Taxonomy Framework: We establish an orthogonal classification scheme covering (1) Scientific Domains and Modalities, (2) Foundation Model Architectures, (3) Physics Integration Levels, and (4) Functional Roles of Reasoning LLMs.
- Systematic Review Protocol: Adhering to PRISMA 2020 standards, every candidate study is screened against strict empirical criteria and validated via official academic API records.
- Synthesis of Evaluation Protocols and Frontiers: We analyze standardized scientific benchmarks (such as WeatherBench 2 and SciTS), map spatial resolution against maximum forecast lead time, and delineate critical open challenges in physical conservation, multi-physics tokenization, and autonomous scientific discovery.

## 2 Background and Preliminaries

### 2.1 Formal Data Modality Formulations

Scientific multimodal temporal datasets exhibit diverse mathematical structures depending on observation platforms, sensing geometries, and sampling frequencies. We formalize the four dominant scientific temporal modalities as follows:

#### 2.1.1 In-Situ Station and Gauge Time Series

Discrete ground monitoring networks (e.g., weather stations, hydrological streamflow gauges, ocean buoys) record continuous physical variables at fixed geospatial coordinates  $(u_i, v_i)$ . An in-situ observation sequence is defined as:

$$\mathbf{X}^{(i)} = \{\mathbf{x}_t^{(i)} \in \mathbb{R}^{C_{\text{station}}}\}_{t=1}^T, \quad (1)$$

where  $T$  represents the temporal horizon and  $C_{\text{station}}$  denotes the recorded physical channels (e.g., precipitation, river discharge, 2-meter air temperature).

#### 2.1.2 Gridded Global/Regional Reanalysis Tensors

Atmospheric and oceanic numerical simulations discretize the planetary sphere  $\mathbb{S}^2$  into a regular latitude-longitude grid with  $H$  latitudinal points and  $W$  longitudinal points across  $L$  vertical vertical pressure levels. The state tensor at time step  $t$  is expressed as:

$$\mathcal{X}_t \in \mathbb{R}^{H \times W \times L \times C_{\text{grid}}}, \quad (2)$$

where  $C_{\text{grid}}$  incorporates upper-air variables (geopotential height  $z$ , temperature  $T$ , zonal and meridional wind components  $u, v$ , specific humidity  $q$ ) alongside surface diagnostic fields (mean sea-level pressure  $msl$ , surface solar radiation).

#### 2.1.3 Multi-Spectral Remote Sensing Cubes

Earth observation satellites (e.g., Sentinel-2, Landsat-8) capture surface reflectance over space, time, and electromagnetic spectral bands. A multi-temporal remote sensing observation cube is represented as:

$$\mathcal{S}_{1:T} \in \mathbb{R}^{T \times B \times H_s \times W_s}, \quad (3)$$

where  $B$  denotes the number of spectral bands (visible RGB, near-infrared, short-wave infrared, synthetic aperture radar polarization), and  $(H_s, W_s)$  denotes the spatial pixel footprint.

#### 2.1.4 High-Frequency Continuous Waveforms

Geophysical and seismological networks deploy seismometers recording ground motion acceleration or velocity across three orthogonal physical axes (Vertical  $Z$ , North-South  $N$ , East-West  $E$ ). A continuous seismic waveform segment is formalized as:

$$\mathbf{W} \in \mathbb{R}^{3 \times N_s}, \quad (4)$$

where  $N_s = f_s \cdot \Delta\tau$  is the number of discrete digital samples collected at sampling frequency  $f_s$  (typically 50–100 Hz).

## 2.2 Core Problem Formulations

### 2.2.1 Deterministic Autoregressive Rollout Forecasting

$$\hat{\mathcal{X}}_{t+1:t+K} = \mathcal{M}_\theta(\mathcal{X}_{t-M+1:t}), \quad (5)$$

optimized via latitude-weighted multi-step rollout loss:

$$\mathcal{L}_{\text{det}} = \sum_{k=1}^K \lambda_k \|\hat{\mathcal{X}}_{t+k} - \mathcal{X}_{t+k}\|_{\mathbf{w}_{\text{lat}}}^2, \quad (6)$$

where  $\|\cdot\|_{\mathbf{w}_{\text{lat}}}$  represents a latitude-weighted norm accounting for spherical grid cell area convergence toward the poles:

$$w_{\text{lat}}(i) = \frac{\cos(\phi_i)}{\frac{1}{H} \sum_{j=1}^H \cos(\phi_j)}. \quad (7)$$

### 2.2.2 Probabilistic Ensemble Generation

Because chaotic atmospheric and oceanic dynamics exhibit sensitivity to initial conditions (the butterfly effect), deterministic point forecasts blur at longer lead times. Probabilistic foundation models aim to sample from the conditional predictive distribution:

$$p_\theta(\mathcal{X}_{t+1:t+K} \mid \mathcal{X}_{t-M+1:t}), \quad (8)$$

minimized using Continuous Ranked Probability Score (CRPS) or diffusion score-matching loss objectives [4].

### 2.2.3 Scientific Question-Answering and Autonomous Reasoning

Let  $\mathcal{Q}$  represent a natural language scientific inquiry, and  $\mathcal{D}_{\text{time}}$  represent an associated temporal or spatio-temporal data sequence. A scientific reasoning LLM  $\mathcal{F}_{\text{LLM}}$  generates explanatory reasoning chains and structured conclusions  $\mathcal{Y}$ :

$$\mathcal{Y} = \mathcal{F}_{\text{LLM}}(\mathcal{Q}, \mathcal{E}(\mathcal{D}_{\text{time}})), \quad (9)$$

where  $\mathcal{E}$  is a modality-specific continuous tokenizer or latent temporal encoder [5].

## 3 Taxonomy and Overview

To systematically classify the rapidly expanding body of literature on scientific multimodal time series foundation models and reasoning LLMs, we construct a four-dimensional taxonomy framework, as illustrated in Fig. 1. Unlike general-purpose temporal surveys that divide models simply by task (e.g., forecasting vs. imputation), our categorization captures the unique physical, mathematical, and algorithmic requirements of scientific domain applications.

### 3.1 Dimension 1: Scientific Domains and Modalities

Scientific time series foundation models specialize across distinct physical environments:

- **Weather and Climate:** Characterized by 3D spherical atmospheric reanalysis fields (ERA5, MERRA-2, IFS HRES), satellite radiometer scans, and ground radar nowcasting sequences [6], [7], [8], [9], [10], [11].
- **Earth Observation and Remote Sensing:** Composed of irregular multi-spectral satellite revisit series (Sentinel-1 SAR, Sentinel-2 Optical, Landsat-8), high-resolution elevation maps, and surface reflectance temporal trajectories [12], [13], [14], [15], [16], [17], [18].

- **Hydrology and Water Resources:** Integrating multi-basin river streamflow gauges, topological river drainage networks, and gridded surface precipitation/evapotranspiration dynamics [19], [20].
- **Oceanography and Marine Systems:** Involving global eddy-resolving reanalysis fields (e.g., GLORYS12V1), sea surface height anomalies, temperature/salinity profiles, and acoustic sensor logs [21], [22], [23].
- **Geophysics and Seismology:** Continuous 3-component seismic waveforms, phase arrival triggers, fault slip observations, and GPS tectonic deformation velocities [24], [25], [26].
- **Space Weather and Heliophysics:** Continuous solar extreme ultraviolet (EUV) magnetograms, solar wind ion velocities, and geomagnetic disturbance indices [27].
- **Polar Cryosphere and Sea Ice Dynamics:** Involving satellite synthetic aperture radar (SAR) swaths, microwave radiometer brightness temperatures, ice-sheet altimetry, and continuous sea ice concentration fields governing planetary thermodynamic albedo [28], [29].
- **Cross-Domain Multimodal Foundations:** Universal time series representations bridging sensor networks, in-situ monitors, and multi-scale physical dynamics [30], [31].

As visualized in the Domain  $\times$  Modality Research Maturity Matrix (Fig. 2), while gridded weather reanalysis and multi-spectral satellite imagery have achieved high maturity, cross-modal integration (e.g., coupling continuous waveforms with satellite imagery or ground gauges with scientific text logs) remains an active frontier.

### 3.2 Dimension 2: Foundation Model Architectures

We categorize foundation backbones by their spatial-temporal inductive biases:

- **3D Spatial and Spherical Transformers:** Employing 3D Earth-specific patch embeddings, hierarchical Swin-like windowed attention, and vertical pressure-level encodings [6], [9], [32], [10].
- **Multi-Mesh Graph Neural Networks:** Discretizing the spherical Earth via refined recursive icosahedral meshes to ensure uniform spatial resolution without polar distortions [7].
- **Continuous Neural Operators:** Leveraging Fourier transforms and spherical harmonic convolutions (e.g., AFNO, SFNO) to learn discretization-invariant operators on Riemannian function spaces [8], [33].
- **High-Order Tensor and Quantum-Classical Hybrid Operators:** Factorizing infinite-dimensional spectral convolution weights via Tucker and Tensor-Train factorizations or embedding spectral transformation into parameterized quantum circuits (PQC) for scalable PDE solving and wavefield tomography [34], [35], [36].
- **Multimodal Masked Autoencoders and Continuous Hypernetworks:** Extending masked autoencoding (MAE) across spatial patches, temporal revisit cycles, and dynamic wavelength hypernetworks [12], [13], [14], [15], [18], [31].
- **Generative Diffusion Models:** Utilizing score-based diffusion denoising to generate calibrated, physically

**Taxonomy Tree: Categorization of Scientific Multimodal Time Series Foundation Models**



Fig. 1. Comprehensive taxonomy framework for scientific multimodal time series foundation models and reasoning LLMs, structured across four orthogonal dimensions: Scientific Domains & Modalities, Foundation Architectures, Physics Integration Levels, and Roles of Reasoning LLMs.

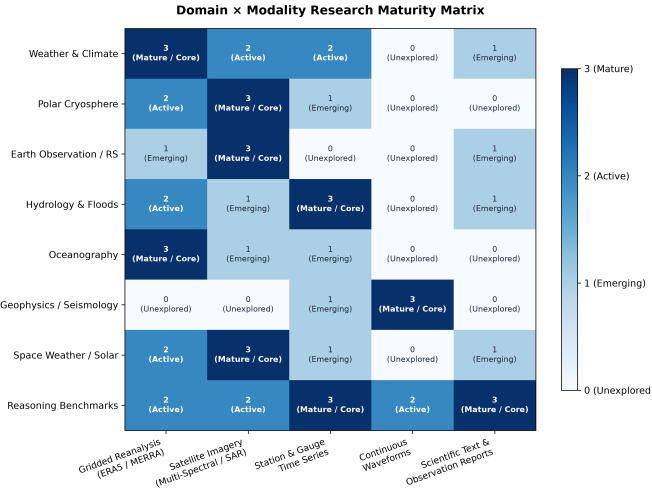


Fig. 2. Domain  $\times$  Modality Research Maturity Matrix, highlighting established core areas and emerging multimodal frontiers.

perturbed probabilistic weather and extreme event ensembles [4], [11], [37].

- Unified Multi-Task Foundational Transformers: Applying prompt-conditioned masking and task tokens across heterogeneous time series topologies [30], [17].

### 3.3 Dimension 3: Physics Integration Levels

The physical fidelity of deep learning models ranges along four distinct levels:

- Purely Data-Driven:** Models optimize statistical reconstruction (e.g., mean squared error, MAE) over historical datasets without explicit differential constraints, relying solely on model capacity to reflect physical dynamics [6], [7], [17].
- Soft Physics Loss Penalties:** Physical conservation principles (e.g., kinetic energy conservation, mass balance, divergence-free constraints, vertical pressure weighting) are formulated as penalty terms within the training loss function [38], [8], [10], [11].
- Hard Architectural Conservation Constraints:** Structural projections (e.g., spherical harmonics basis expansion, divergence-free projection layers) guarantee that model predictions strictly satisfy conservation laws by construction [33].
- Hybrid PDE-Neural Solvers:** Deep neural representations parameterize unresolved subgrid physical processes (convection, clouds, turbulence) coupled directly to differentiable numerical fluid dynamical cores [39], [40], [19].

### 3.4 Dimension 4: Functional Roles of Reasoning LLMs

Large Language Models engage with scientific temporal data through four operational archetypes:

- **Scientific Reasoner:** Understanding physical time series tokens and generating natural language explanations, causal attribution, and hypothesis evaluation across scientific domains [5], [41], [42].
- **Contextual Enhancer:** Supplying auxiliary domain semantics, station metadata, and climate zone priors to guide numerical prediction backbones [43], [31].
- **Autonomous Tool-Using Agent:** Decomposing complex scientific inquiries, retrieving real-time sensor streams, and executing scientific diagnostic libraries (e.g., CDO, xarray, SciPy) to produce verified scientific analyses [44], [23].
- **Interactive Natural Language Interface:** Translating unstructured scientist inquiries into executable database queries and simulation configurations.

## 4 Core Foundation Model Architectures

### 4.1 3D Spatial and Spherical Transformers

Traditional 2D Vision Transformers process planar images by dividing them into uniform patches. However, planetary atmospheric states are inherently three-dimensional, structured across latitude, longitude, and geopotential pressure altitudes.

#### 4.1.1 Pangu-Weather: 3D Earth-Specific Transformer

Bi et al. [6] introduced Pangu-Weather, the first deep learning weather forecasting system to demonstrably surpass top-tier operational NWP models (ECMWF IFS) in medium-range deterministic forecasting accuracy. Pangu-Weather formulates atmospheric prediction using a 3D Earth-Specific Transformer (3DEST) architecture. The state space is divided into 3D cuboid patches of size  $2 \times 4 \times 4$  across vertical levels, latitude, and longitude. To account for inhomogeneous spatial dynamics (e.g., Coriolis force variations across latitudes and atmospheric lapse rates across altitudes), 3DEST replaces standard relative position biases with an Earth-Specific Positional Bias (ESPB) matrix indexed explicitly by pressure level and latitude:

$$\text{Attn}(Q, K, V) = \text{Softmax} \left( \frac{QK^T}{\sqrt{d}} + B_{\text{Earth}}(l_i, l_j, \phi_i, \phi_j) \right) V. \quad (10)$$

To counteract rapid error accumulation during iterative autoregressive rollout, Pangu-Weather trains four distinct models with lead times of 1, 3, 6, and 24 hours. A greedy temporal aggregation algorithm chains these models to generate 7-day global forecasts in seconds.

#### 4.1.2 FuXi: Cascade Forecasting System

Recognizing that predictive uncertainty compounds nonlinearly with lead time, Chen et al. [32] developed FuXi, a cascade machine learning system consisting of three specialized Swin Transformer stages: FuXi-Short (0–42 hours), FuXi-Medium (42–120 hours), and FuXi-Long (120–360 hours). By dynamically switching between model weights optimized for different forecasting horizons, FuXi significantly curtails the blurring and spectral dissipation common in long-horizon neural forecasts.

### 4.1.3 Aurora: Scaling to Planetary Foundation Models

Bodnar et al. [9] demonstrated that scaling principles hold in Earth system modeling by introducing Aurora, a 1.3-billion parameter foundation model. Pretrained on over 1 petabyte of heterogeneous atmospheric data—including ERA5, ECMWF HRES, operational IFS analyses, and NOAA GFS—Aurora incorporates a 3D Perceiver architecture capable of accepting arbitrary combinations of vertical levels and surface variables at  $0.1^\circ$  operational resolution (10 km equivalent).

#### 4.1.4 AIFS: ECMWF Operational GNN-Transformer Architecture

Lang et al. [45] developed the Artificial Intelligence Forecasting System (AIFS), ECMWF’s official operational data-driven model. AIFS marries the geometric flexibility of graph neural networks with the long-range receptive fields of sliding-window transformers. The architecture maps icosahedral multi-mesh node representations through a graph encoder, processes latent multi-scale interactions via a local sliding-window transformer that limits attention memory to  $O(N \cdot W)$  tokens, and decodes back to the operational reduced Gaussian grid via a graph decoder. Pretrained on decades of ERA5 reanalysis and fine-tuned on ECMWF’s operational NWP analyses, AIFS runs in operational production at ECMWF four times daily alongside physics-based NWP systems.

## 4.2 Multi-Mesh Spherical Graph Neural Networks

### 4.2.1 GraphCast: Planetary Multi-Mesh Routing

Lam et al. [7] pioneered an alternative geometric approach by formulating global weather forecasting over a multi-resolution spherical graph. Standard equirectangular projections concentrate excessive grid cells near the poles while diluting them at the equator. GraphCast circumvents this singularity using an Encode-Process-Decode framework operating on a hierarchical icosahedral mesh:

- 1) **Grid-to-Mesh Encoder:** Maps  $0.25^\circ$  latitude-longitude grid variables onto the vertices of a 6-level refined icosahedral mesh  $M_6$  (40,962 nodes, 122,880 spherical edges).
- 2) **Multi-Mesh Processor:** Employs 16 layers of homogeneous message passing across all hierarchical mesh levels ( $M_0$  to  $M_6$ ). Long-range edges in coarse mesh levels ( $M_0, M_1$ ) facilitate rapid, global information propagation across planetary teleconnection pathways in few message-passing hops.
- 3) **Mesh-to-Grid Decoder:** Projects refined mesh vertex features back to the  $0.25^\circ$  target grid to predict single-step 6-hour temporal increments  $\Delta \mathcal{X}_t$ .

GraphCast is trained with a 12-step autoregressive loss rollout, enabling robust, physically stable 10-day medium-range forecasts that outperform ECMWF HRES on over 90% of headline verification variables.

## 4.3 Continuous Neural Operators

### 4.3.1 FourCastNet: Adaptive Fourier Neural Operators

Pathak et al. [8] deployed Adaptive Fourier Neural Operators (AFNO) to solve planetary-scale atmospheric PDEs. Conventional spatial convolutions are intrinsically tied to fixed

grid resolutions. In contrast, neural operators approximate non-linear mappings between infinite-dimensional function spaces. AFNO transforms tokenized feature representations into the frequency domain via 2D Fast Fourier Transforms (FFT), applies block-diagonal weight multiplications with soft-thresholding sparsity in the frequency domain, and returns to physical space via inverse FFT:

$$z_{\text{out}}(x) = \mathcal{F}^{-1}(\text{SoftThreshold}(\mathbf{W} \cdot \mathcal{F}(z_{\text{in}})(k))) + \mathbf{W}_s z_{\text{in}}(x). \quad (11)$$

This formulation provides computational complexity quasi-linear in spatial resolution while demonstrating remarkable zero-shot super-resolution and transferability.

#### 4.3.2 SFNO: Spherical Fourier Neural Operators

While planar Fourier transforms accelerate spectral computation, standard 2D FFTs applied to equirectangular latitude-longitude coordinates suffer from severe metric distortions near the poles, where meridians converge. Bonev et al. [33] resolved this geometric bottleneck by formulating the Spherical Fourier Neural Operator (SFNO), which solves continuous PDEs directly on the Riemannian manifold of the unit sphere  $\mathbb{S}^2 \subset \mathbb{R}^3$ .

Let colatitude be  $\theta \in [0, \pi]$  and longitude be  $\lambda \in [0, 2\pi)$ . Any square-integrable scalar atmospheric state  $u \in \mathcal{L}^2(\mathbb{S}^2)$  admits an orthogonal expansion in terms of spherical harmonics  $Y_\ell^m(\theta, \lambda)$ :

$$u(\theta, \lambda) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \hat{u}_\ell^m Y_\ell^m(\theta, \lambda), \quad (12)$$

where  $\ell \in \mathbb{N}_0$  denotes the spherical degree,  $m \in \{-\ell, \dots, \ell\}$  is the order, and the orthonormal basis functions are defined via associated Legendre polynomials  $P_\ell^m$ :

$$Y_\ell^m(\theta, \lambda) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\lambda}. \quad (13)$$

Continuous convolution on  $\mathbb{S}^2$  with an invariant kernel  $\kappa$  is realized through spherical harmonic transforms  $\mathcal{SHT}$  and inverse transforms  $\mathcal{ISHT}$ :

$$(\kappa \star_{\mathbb{S}^2} u)(\theta, \lambda) = \mathcal{ISHT}(\mathbf{W}_\ell \cdot \mathcal{SHT}(u)(\ell, m)), \quad (14)$$

where  $\mathbf{W}_\ell \in \mathbb{C}^{d_{\text{out}} \times d_{\text{in}}}$  denotes degree-dependent weight tensors. By the convolution theorem on  $\text{SO}(3)$ , this spherical harmonic formulation provides exact rotational equivariance, eliminates high-latitude numerical artifacts, and ensures long-term spectral stability over multi-day atmospheric rollouts.

#### 4.4 Multimodal Masked Autoencoders and Latent Alignments

In the remote sensing and Earth observation domains, observations arrive as asynchronous, multi-spectral satellite imagery cubes, irregular in-situ sensor readings, and gridded reanalysis tensors.

- SatMAE [12] introduced a temporal and multi-spectral Vision Transformer trained via masked autoencoding. By decoupling spatial, temporal, and spectral positional embeddings, SatMAE processes multi-temporal revisit sequences with high masking ratios (up to 75%),

outperforming supervised baselines on land cover and crop yield estimation.

- Presto [13] developed a lightweight (1.2M parameters) transformer tailored for pixel-level time series. Presto natively ingests heterogeneous modalities—including Sentinel-1 SAR backscatter, Sentinel-2 multi-spectral bands, elevation, and ERA5 weather—yielding exceptional few-shot transfer performance.
- Galileo [14] expanded multi-sensor pretraining across spatial-temporal scales using dual global and local contrastive objectives, enabling unified geospatial segmentation and dynamic flood mapping.
- SkySense [46] established a billion-scale (1.9B parameter) multi-modal remote sensing foundation model. By formulating a factorized spatio-temporal cross-attention architecture, SkySense jointly pretrains on over 21.5 million multi-modal satellite temporal sequences (combining optical multi-spectral reflectance with synthetic aperture radar SAR backscatter). It establishes state-of-the-art generalist performance across 16 diverse Earth observation benchmarks, including fine-grained crop classification, building damage assessment, and temporal change detection.
- DOFA [18] introduced a Dynamic One-For-All foundation architecture inspired by neural plasticity. Recognizing that Earth observation platforms possess disparate band wavelengths and ground sampling distances (GSD), DOFA conditions dynamic weight hypernetworks directly on continuous physical wavelengths  $\lambda \in [400 \text{ nm}, 14,000 \text{ nm}]$ , unifying optical, SAR, and hyperspectral temporal modalities under a single pre-trained representation.
- GEO-Bench [47] established a standardized benchmark suite for Earth monitoring foundation models. Unifying multi-temporal Sentinel-1/2, SPOT, and m-CASIS data across regional land-use segmentation and multi-spectral classification, GEO-Bench provides rigorous zero-shot and fine-tuning evaluation protocols for multi-sensor satellite representations.
- EarthPT [17] developed a 700M-parameter autoregressive decoder foundation model specifically designed for Earth Observation time series. Treating multi-spectral reflectance measurements (400–2300 nm) across Copernicus Sentinel-2 pixel-year revisit sequences as dense temporal tokens, EarthPT predicts future surface reflectance evolutions with high physical fidelity.
- Prithvi [15] and its weather-climate counterpart Prithvi WxC [48] established open-source geospatial foundation backbones jointly developed by NASA and IBM, scaling to 2.3 billion parameters.

##### 4.4.1 Formal Alignment Paradigms across Heterogeneous Geometries

As illustrated in Fig. 3, cross-modal integration across disparate spatial geometries follows three dominant architectural paradigms:

- 1) Strategy A: Early Heterogeneous Patch Projection: Given heterogeneous modality inputs  $\mathbf{X}^{(m)} \in \mathbb{R}^{T_m \times C_m \times H_m \times W_m}$  with disparate channel definitions

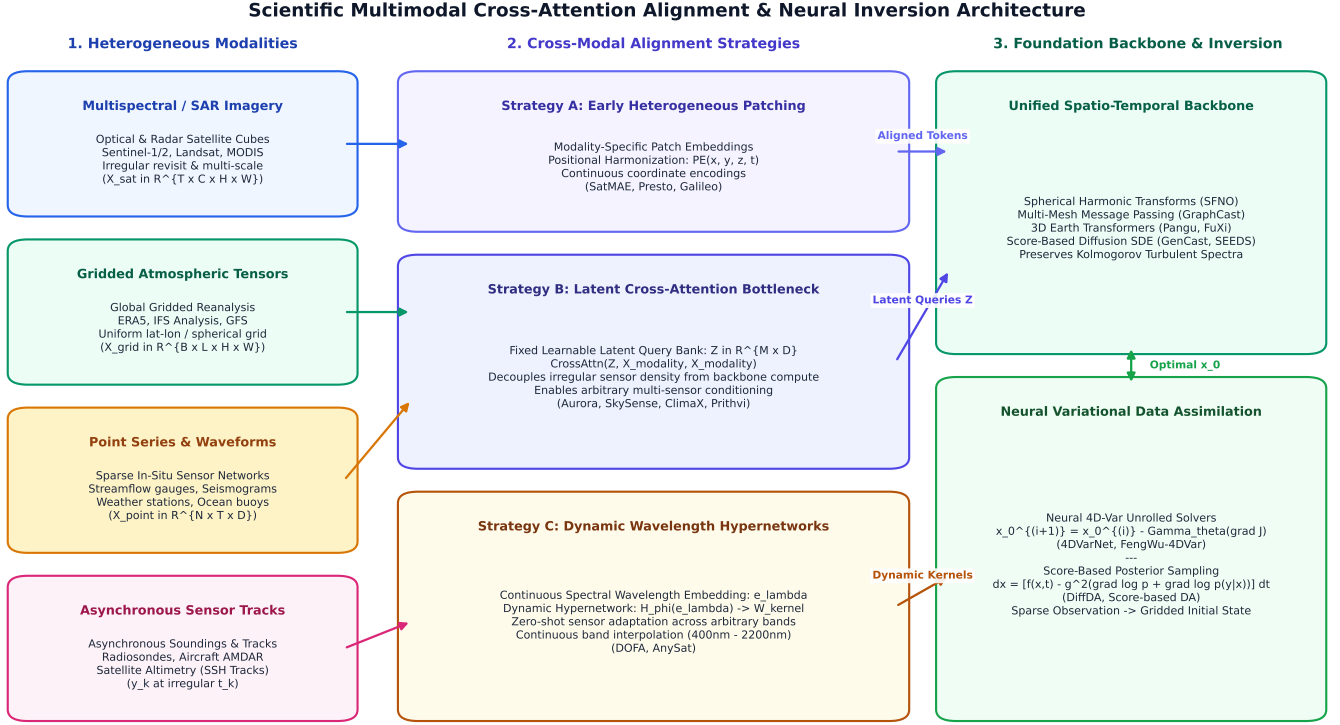


Fig. 3. Architectural taxonomy of scientific multimodal cross-attention alignment and neural variational data assimilation across heterogeneous spatial geometries. Left: Multi-source observational modalities spanning satellites, reanalyses, point sensor networks, and asynchronous tracks. Center: Three primary alignment paradigms (Early Heterogeneous Patching, Latent Cross-Attention Bottlenecks, and Dynamic Wavelength Hypernetworks). Right: Unified spatio-temporal foundation backbones and neural variational data assimilation solvers bridging sparse observation streams directly into gridded initial state representations.

and Ground Sampling Distances (GSD), modality-specific linear or convolutional patch embedding kernels  $\mathbf{E}^{(m)}$  project local spatial-temporal patches into a shared hidden dimension  $D$ :

$$\mathbf{h}_i^{(m)} = \mathbf{E}^{(m)} \mathbf{x}_i^{(m)} + \mathbf{PE}(x_i, y_i, z_i, t_i), \quad (15)$$

where  $\mathbf{PE}(\cdot)$  denotes a unified continuous 4D Fourier spatio-temporal positional encoding, ensuring coordinate harmonization across disparate satellite swaths and ground station coordinates (utilized in SatMAE [12] and Presto [13]).

- 2) Strategy B: Latent Cross-Attention Bottlenecks: When observation token counts  $N_{obs} = \sum_m N_m$  fluctuate dramatically due to dynamic sensor availability or dense planetary grids, feeding all tokens directly into full self-attention scales quadratically as  $O(N_{obs}^2)$ . To circumvent this computational explosion, Perceiver-style architectures (Aurora [9], SkySense [46]) introduce a fixed bank of learnable latent query tokens  $\mathbf{Z} \in \mathbb{R}^{M \times D}$  ( $M \ll N_{obs}$ ) that attend to the irregular multi-sensor input tokens via cross-attention:

$$\mathbf{Z}_{aligned} = \text{Softmax} \left( \frac{\mathbf{Z} \mathbf{W}_Q (\mathbf{H}_{obs} \mathbf{W}_K)^T}{\sqrt{d}} \right) (\mathbf{H}_{obs} \mathbf{W}_V), \quad (16)$$

where  $\mathbf{H}_{obs} = [\mathbf{H}^{(1)}; \dots; \mathbf{H}^{(K)}]$  concatenates all available sensor tokens. The spatio-temporal foundation backbone then processes only the constant-

size latent state  $\mathbf{Z}_{aligned}$  with linear complexity in input resolution, decoupling observation density from computation.

- 3) Strategy C: Dynamic Wavelength Hypernetworks: To support arbitrary spectral bands without modifying network weights, continuous wavelength hypernetworks (DOFA [18]) encode the physical central wavelength  $\lambda$  of each spectral channel via harmonic Fourier embeddings  $\mathbf{e}_\lambda = [\sin(2^k \pi \lambda), \cos(2^k \pi \lambda)]_{k=0}^{K-1}$ . A lightweight Multi-Layer Perceptron (MLP) hypernetwork  $\mathcal{H}_\phi$  maps  $\mathbf{e}_\lambda$  to generate the convolution or linear projection weights dynamically:

$$\mathbf{W}_{patch}(\lambda) = \mathcal{H}_\phi(\mathbf{e}_\lambda) \in \mathbb{R}^{D \times (P^2 \cdot C_{in})}. \quad (17)$$

This enables zero-shot sensor transfer across optical (e.g., Sentinel-2 10m/20m bands), thermal infrared (Landsat-8/TIRS), and radar backscatter (Sentinel-1 VV/VH) modalities within a unified model.

#### 4.5 Neural Data Assimilation and Sensor-to-Grid Variational Inversion

A fundamental gap separating operational numerical weather forecasting from purely data-driven foundation models is data assimilation (DA): the mathematical process of synthesizing sparse, asynchronous, and noisy physical observations into a physically consistent gridded initial atmospheric or oceanic

state  $\mathbf{x}_0$ . While existing foundation models (e.g., Pangu-Weather, GraphCast) rely entirely on pre-computed gridded reanalyses (ERA5 or operational IFS analyses) produced by supercomputing NWP centers, emerging architectures introduce differentiable variational and score-based neural assimilation solvers to invert raw sensor streams directly into gridded model initial states.

#### 4.5.1 Variational Formulation and Neural 4D-Var

In classical numerical weather prediction, four-dimensional variational data assimilation (4D-Var) optimizes an initial state  $\mathbf{x}_0$  over an assimilation window  $[t_0, t_K]$  to minimize a penalized least-squares cost function:

$$\mathcal{J}(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \sum_{k=0}^K \|\mathbf{y}_k - \mathcal{H}_k(\mathcal{M}_{0 \rightarrow k}(\mathbf{x}_0))\|_{\mathbf{R}_k^{-1}}^2, \quad (18)$$

where  $\mathbf{x}_b$  is the background prior state,  $\mathbf{B}$  is the background error covariance matrix,  $\mathbf{y}_k$  denotes sparse observations at time  $t_k$ ,  $\mathbf{R}_k$  is the observation error covariance,  $\mathcal{H}_k$  is the observation forward operator, and  $\mathcal{M}_{0 \rightarrow k}$  is the non-linear dynamical forward model. In operational NWP, minimizing Eq. (18) requires maintaining complex adjoint code  $\mathcal{M}^*$  to compute gradients  $\nabla_{\mathbf{x}_0} \mathcal{J}$ , consuming up to 80% of operational supercomputing compute budgets.

To overcome this bottleneck, Fablet et al. [49] introduced 4DVarNet, reformulating variational data assimilation as an end-to-end learnable gradient-based solver. 4DVarNet replaces the analytical minimization loop with an unrolled recurrent neural architecture:

$$\mathbf{x}_0^{(i+1)} = \mathbf{x}_0^{(i)} - \mathbf{\Gamma}_\theta \left( \nabla_{\mathbf{x}_0} \mathcal{J}(\mathbf{x}_0^{(i)}), \mathbf{h}^{(i)} \right), \quad (19)$$

where  $\mathbf{\Gamma}_\theta$  is a learned recurrent neural operator (e.g., ConvLSTM or U-Net solver),  $\mathbf{h}^{(i)}$  represents internal solver memory states, and the gradient  $\nabla_{\mathbf{x}_0} \mathcal{J}$  is computed via automated differentiation through the observation operator  $\mathcal{H}_k$  and neural forward dynamics  $\mathcal{M}_\phi$ . Originally deployed for sea surface height (SSH) reconstruction from sparse satellite altimetry tracks in oceanography, 4DVarNet achieves orders-of-magnitude faster convergence than classical conjugate gradient descent while learning dynamic background error covariances directly from historical data.

Scaling neural data assimilation to global atmospheric forecasting, Xiao et al. [50] introduced FengWu-4DVar. FengWu-4DVar couples the FengWu multi-scale atmospheric transformer with an unrolled 4D variational optimization loop. By exploiting the end-to-end differentiability of the neural forecasting backbone, FengWu-4DVar directly back-propagates observation residuals from asynchronous in-situ soundings and satellite radiances through 24-hour assimilation windows, generating initialized atmospheric states that match operational ECMWF IFS analysis quality while reducing assimilation wall-clock time from hours to seconds.

#### 4.5.2 Score-Based and Diffusion Data Assimilation

Complementing deterministic variational solvers, generative diffusion models reframe data assimilation as conditional posterior sampling: drawing physical state trajectories  $\mathbf{x}_{0:K} \sim p(\mathbf{x}_{0:K} | \mathbf{y}_{0:K})$ .

Rozet and Louppe [51] formulated Score-based Data Assimilation (SDA), showing that continuous score-based

stochastic differential equations can execute exact spatio-temporal smoothing and filtering over non-linear fluid dynamics without tangent-linear approximations. Given a trained unconditional foundation prior  $s_\theta(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ , SDA conditions the reverse-time generative SDE on sparse partial observations  $\mathbf{y}$  via Tweedie’s guidance:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g(t)^2 (\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t))] dt + g(t) d\mathbf{w} \quad (20)$$

Under Gaussian observation noise  $\mathbf{y} \sim \mathcal{N}(\mathbf{H}\mathbf{x}_0, \mathbf{R})$ , the intractable conditional score  $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)$  is approximated analytically via the posterior expectation  $\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t))$ :

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t) \approx \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{H}^T (\mathbf{H} \Sigma_t \mathbf{H}^T + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_0(\mathbf{x}_t)). \quad (21)$$

Scaling diffusion assimilation to weather-scale operational state estimation, Huang et al. [52] introduced DiffDA. DiffDA trains a conditional denoising diffusion probabilistic model (DDPM) conditioned on sparse surface weather station observations and radiosonde soundings. Operating directly on physical observation coordinates, DiffDA samples ensemble analyses that accurately reproduce fine-scale frontal boundaries, atmospheric rivers, and convective storms, demonstrating that neural generative assimilation can bridge heterogeneous sensor observations directly into gridded foundation state spaces.

#### 4.6 Probabilistic Diffusion and Extreme-Event Ensembles

A pervasive theoretical limitation of deterministic forecasting models trained under mean squared error ( $L_2$ ) or mean absolute error ( $L_1$ ) losses is their vulnerability to spectral blurring and variance dissipation at extended forecast horizons. Formally, given an observation history  $\mathbf{y}_{1:t}$  and future atmospheric target  $\mathbf{x}_{t+\tau}$ , optimizing an  $L_2$  regression objective  $\min_\theta \mathbb{E}[\|\mathbf{x}_{t+\tau} - f_\theta(\mathbf{y}_{1:t})\|^2]$  provably converges to the conditional posterior mean:

$$f_\theta^*(\mathbf{y}_{1:t}) = \mathbb{E}[\mathbf{x}_{t+\tau} | \mathbf{y}_{1:t}] = \int \mathbf{x} p(\mathbf{x} | \mathbf{y}_{1:t}) d\mathbf{x}. \quad (22)$$

Because atmospheric flow exhibits non-linear chaotic dynamics with positive Lyapunov exponents, the true posterior probability density  $p(\mathbf{x} | \mathbf{y}_{1:t})$  bifurcates into multi-modal trajectories at extended lead times ( $\tau \geq 5$  days). Averaging over these distinct physical realizations washes out high-wavenumber turbulent eddies, severely dampens the kinetic energy power spectrum  $E(k) \propto k^{-5/3}$  or  $k^{-3}$ , and underestimates extreme-event tail probabilities  $P(\|\mathbf{x}\| > \tau_{\text{threshold}})$ .

To resolve this limitation, generative diffusion models reframe weather forecasting as stochastic sampling from the true conditional distribution  $p(\mathbf{x} | \mathbf{y})$ .

##### 4.6.1 Continuous Score-Based Diffusion Formulation

Let the atmospheric state trajectory follow a continuous forward stochastic differential equation (SDE) over diffusion time  $s \in [0, 1]$ :

$$d\mathbf{x}_s = \mathbf{f}(\mathbf{x}_s, s) ds + g(s) d\mathbf{w}, \quad (23)$$

where  $\mathbf{w}$  is standard Brownian motion,  $\mathbf{f}$  represents drift, and  $g(s)$  governs noise dispersion. By Anderson’s reverse-time theorem, the generative reverse trajectory is governed by:

$$d\mathbf{x}_s = [\mathbf{f}(\mathbf{x}_s, s) - g(s)^2 \nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s | \mathbf{y})] ds + g(s) d\bar{\mathbf{w}}, \quad (24)$$

where  $d\bar{\mathbf{w}}$  denotes backward Brownian motion, and the score function  $s_\theta(\mathbf{x}_s, s, \mathbf{y}) \approx \nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s | \mathbf{y})$  is parameterized by a deep neural operator or graph network trained via denoising score matching.

#### 4.6.2 GenCast, SEEDS, and FuXi-Extreme: Operational Diffusion Ensembling

Price et al. [4] operationalized this paradigm in GenCast, conditioning a score-matching network on spherical multi-mesh graphs using historical 6-hour atmospheric increments  $(\mathbf{x}_{t-6h}, \mathbf{x}_t)$ . By sampling 50 stochastic trajectories, GenCast accurately generates medium-range ensemble forecasts that maintain realistic energy spectra up to Day 15, outperforming ECMWF ENS in Continuous Ranked Probability Scores (CRPS) across over 80% of headline atmospheric variables and drastically improving tropical cyclone track and intensity predictions.

Concurrently, Li et al. [11] introduced SEEDS (Scalable Ensemble Envelope Diffusion Sampler), demonstrating that generative diffusion can emulate operational numerical weather forecast ensembles (such as NOAA GEFS) conditioned on only 1 or 2 deterministic forecast trajectories. SEEDS generates large ensembles (e.g., 50 members) at  $1^\circ$  operational resolution orders of magnitude faster than numerical ensembles, precisely reproducing the spread-skill relationship and capturing extreme heatwave and storm tail events without numerical energy collapse.

Extending diffusion specifically to hazard extremes, Zhong et al. [53] formulated FuXi-Extreme. While deterministic weather models drastically underestimate tail precipitation rates and gale-force wind gusts, FuXi-Extreme couples a cascade Swin Transformer with a conditional latent diffusion model. Trained on high-resolution ERA5 extreme events, it generates realistic sharp spatial gradients for extreme precipitation ( $> 50$  mm/day) and 10-meter wind gusts, resolving the longstanding spectral smoothing dilemma for operational disaster early warnings.

### 4.7 Universal Tokenization and Cross-Domain Temporal Transferability

A major architectural obstacle in scientific foundation models is the profound heterogeneity of physical data representations: regular lat-lon reanalysis grids (ERA5), irregular unstructured meshes (finite-volume ocean models), sparse in-situ station networks (hydrology streamflow gauges), 1D continuous high-frequency waveforms (seismology), and irregular multi-spectral satellite revisits (remote sensing). Recent research develops universal tokenization mechanisms to bridge this divide:

#### 4.7.1 Unified Multi-Task Architectures: UniTS

Gao et al. [30] developed UniTS, a unified multi-task foundation model for time series. Unlike domain-specific architectures requiring separate parameterizations for each dataset,

UniTS processes arbitrary numbers of variables and sequences by introducing:

- 1) Sequence-Variable Tokenization: Decouples temporal sequence patches and variable channels into modular tokens via lightweight linear projections.
- 2) Prompt-Conditioned Task Tokens: Ingests task specification tokens (e.g., forecasting horizon, classification labels, imputation mask) into self-attention layers, conditioning a single shared transformer backbone to execute forecasting, anomaly detection, imputation, and classification across 38 diverse datasets without architectural modification.

#### 4.7.2 Language Model Tokenization for Time Series: Chronos

Ansari et al. [54] introduced Chronos, demonstrating that classical language model tokenization principles translate directly to continuous scientific and physical time series. Chronos discretizes real-valued time series through affine mean-variance normalization followed by uniform quantile binning into a fixed vocabulary of  $V$  discrete tokens (e.g., 4096 bins). Pretrained on diverse time-series corpora using standard cross-entropy auto-regression over T5 architectures, Chronos demonstrates that language model pretraining objectives yield remarkable zero-shot probabilistic forecasting capabilities across diverse physical and socio-economic systems.

#### 4.7.3 Autoregressive Probabilistic Foundations: Lag-Llama

Rasul et al. [55] formulated Lag-Llama, a dedicated probabilistic foundation model utilizing a decoder-only LLaMA-style transformer architecture. Ingesting historical lagged temporal features and tokenized continuous values, Lag-Llama applies rotary position embeddings (RoPE) and outputs parametric Student- $t$  distribution parameters  $(\mu, \sigma, \nu)$  at each step. This formulation natively captures fat-tailed physical distributions and extreme deviations across physical sensor streams.

#### 4.7.4 Open Foundation Models: MOMENT

Goswami et al. [31] introduced MOMENT, an open-source family of foundation models trained on the Time-series Pile, a curated corpus of over 13 million diverse sequences spanning earth science sensors, industrial machinery, and physical processes. Using a patched T5-style masked autoencoder architecture, MOMENT establishes consistent zero-shot and few-shot transfer capabilities across diverse scientific temporal benchmarks.

#### 4.7.5 Atmospheric Patch Randomization: Stormer

Nguyen et al. [10] introduced Stormer, demonstrating that a standard Vision Transformer backbone can match or exceed specialized weather forecasting architectures when coupled with two critical physical inductive biases:

- 1) Randomized Dynamics Objective: During pretraining, Stormer forecasts atmospheric states across randomized lead times  $\Delta t \sim \mathcal{U}(6h, 120h)$ , forcing the model to learn multi-scale continuous temporal dynamics rather than overfitting to fixed autoregressive step sizes.

- 2) Pressure-Weighted Vertical Loss: Weighs atmospheric layer losses proportional to geopotential layer thickness and density, aligning optimization with physical mass distribution.

#### 4.7.6 Standardized Waveform Tokenization: SeisBench

In geophysics, Woollam et al. [26] established SeisBench, a standardized framework and benchmarking suite unifying 3-component continuous seismic waveform datasets (ETHZ, GEOFON, INSTANCE, SCEDC, STEAD, comprising over 30 million waveforms). By standardizing sampling rates (100 Hz), coordinate rotations, and trace tokenization, SeisBench enables cross-regional foundation pretraining and transferability for phase arrival picking, earthquake detection, and fault rupture localization.

### 4.8 Physical Conservation Constraints and Differentiable Hybrid Solvers

A persistent vulnerability of purely data-driven black-box neural networks is their tendency to violate fundamental physical conservation principles over prolonged autoregressive rollouts, leading to unphysical mass dissipation, kinetic energy damping, or moisture divergence. Recent advances address this through distinct levels of physical integration:

- 1) Soft Physics Regularization: Incorporates auxiliary loss penalties that penalize physical inconsistencies during training:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{conservation}}, \quad (25)$$

where  $\mathcal{L}_{\text{conservation}}$  penalizes deviations in global total dry mass, vertical geostrophic balance, or spectral kinetic energy distributions (employed in FourCastNet [8], FengWu [38], and Stormer [10]).

- 2) Hard Architectural Invariance: Enforces conservation laws and symmetry directly into neural layer transformations. For instance, SFNO [33] guarantees  $\text{SO}(3)$  rotational equivariance on the unit sphere  $\mathbb{S}^2$ , preventing polar singularities and high-latitude numerical distortions by construction.
- 3) Hybrid Multiscale Subgrid Parameterization: ClimSim: Yu et al. [40] established ClimSim, the largest multiscale dataset and benchmark for training deep learning subgrid physics emulators. In multiscale climate modeling (e.g., E3SM-MMF), macro-scale fluid dynamics are integrated by numerical PDE solvers on coarse grids, while complex micro-scale processes (moist convection, cloud formation, turbulence, and radiative transfer) are parameterized by neural networks. ClimSim provides 5.7 billion high-resolution atmospheric column profiles, benchmarking physics-informed ML architectures on strict conservation of total column water mass and dry static energy.
- 4) Differentiable Hybrid Solvers: NeuralGCM: Kochkov et al. [39] coupled a fully differentiable atmospheric dynamical core with neural network parameterizations. The resolved large-scale hydrodynamics are integrated explicitly by an analytical spectral PDE solver that strictly enforces conservation of mass, momentum, and moisture:

$$\frac{\partial \mathbf{X}}{\partial t} = \mathcal{F}_{\text{dynamical}}(\mathbf{X}) + \mathcal{N}_{\theta}(\mathbf{X}), \quad (26)$$

where  $\mathcal{F}_{\text{dynamical}}$  executes analytical fluid dynamics, while the neural network  $\mathcal{N}_{\theta}$  parameterizes sub-grid physical processes. NeuralGCM demonstrates unprecedented multi-decadal stability without climate drift, matching the operational deterministic accuracy of ECMWF IFS HRES at 1–10 day weather horizons while preserving global physical conservation laws for decadal climate simulations.

- 5) Continuous Physics-Informed Neural ODE Dynamics: ClimODE: Verma et al. [56] pioneered ClimODE, fundamentally rethinking scientific time series modeling by replacing discrete multi-step mapping  $\mathbf{x}_{t+\Delta t} = \mathcal{M}_{\theta}(\mathbf{x}_t)$  with continuous differential flow trajectories grounded in fluid mechanics. By explicitly embedding continuity and advective constraints into neural ordinary differential equations, ClimODE achieves global weather forecasting with only 0.3M parameters—over two orders of magnitude smaller than discrete vision transformers—while natively supporting arbitrary continuous-time evaluation intervals.

### 4.9 Continuous-Discrete Physics Invariants, Lie Symmetries, and Geometric Deep Learning

A central mathematical frontier in scientific foundation models is reconciling discrete computational architectures (e.g., attention over tokenized patches, autoregressive discrete rollouts) with the continuous, symmetry-preserving nature of physical laws governing the Earth system.

#### 4.9.1 Continuous Fluid Advection and Neural Differential Flows

Classical discrete autoregressive architectures forecast atmospheric states via discrete recurrence:

$$\mathbf{x}_{k+1} = \mathcal{M}_{\theta}(\mathbf{x}_k), \quad k \in \{0, 1, \dots, K\}, \quad (27)$$

where  $\Delta t = t_{k+1} - t_k$  is strictly locked to a predefined sampling cadence (e.g., 6h in ERA5). This discrete formulation suffers from two fundamental limitations: (1) time-step rigidity, precluding direct assimilation or forecasting at arbitrary asynchronous timestamps, and (2) compounding discretization error, where unconstrained discrete updates accumulate unphysical drift.

In contrast, continuous-time formulation models the planetary atmosphere as a non-linear dynamical system evolving over the continuous manifold  $\mathcal{M} = \mathbb{S}^2 \times \mathbb{R}^+$ :

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}_{\theta}(\mathbf{x}(t), t), \quad \mathbf{x}(t) = \mathbf{x}(t_0) + \int_{t_0}^t \mathbf{f}_{\theta}(\mathbf{x}(\tau), \tau) d\tau. \quad (28)$$

To enforce physical consistency, ClimODE [56] constrains the learned vector field  $\mathbf{f}_{\theta}$  using the Navier–Stokes transport theorem. The continuous conservation of atmospheric mass density  $\rho$  is governed by the continuity PDE:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}_{\theta}) = 0, \quad (29)$$

where  $\mathbf{v}_{\theta}(\mathbf{s}, t) \in T_{\mathbf{s}}\mathbb{S}^2$  is the neural velocity field over spherical coordinates  $\mathbf{s} = (\lambda, \theta)$ . Similarly, any scalar tracer field  $\phi \in \{T, q, Z\}$  (temperature, specific humidity, geopotential) satisfies the continuous material advection-diffusion relation:

$$\frac{D\phi}{Dt} \equiv \frac{\partial \phi}{\partial t} + (\mathbf{v}_{\theta} \cdot \nabla)\phi = \mathcal{D}_{\theta}(\phi) + \mathcal{S}_{\theta}(\phi, \mathbf{s}, t), \quad (30)$$

where  $\mathcal{D}_\theta$  denotes turbulent sub-grid diffusion and  $\mathcal{S}_\theta$  encapsulates radiative source/sink fluxes. Integrating Eq. (28) with adaptive-step numerical ODE solvers (e.g., Dormand–Prince 5(4) or adaptive Runge–Kutta) guarantees that the flow remains bounded along physical trajectories, yielding superior long-term stability and eliminating the need for empirical post-hoc smoothing.

#### 4.9.2 Lie Group Symmetries and Spherical Coordinate Invariance

The fundamental equations of geophysical fluid dynamics (primitive equations, shallow-water equations) are invariant under the continuous Lie group of spatial rotations  $\text{SO}(3)$  and time translation  $\mathbb{R}$ . Let  $g \in \text{SO}(3)$  represent an arbitrary rigid rotation of the planetary sphere  $\mathbb{S}^2$ . A scientific foundation operator  $\mathcal{F}_\theta : \mathcal{X}(\mathbb{S}^2) \rightarrow \mathcal{X}(\mathbb{S}^2)$  is said to be  $\text{SO}(3)$ -equivariant if and only if:

$$\mathcal{F}_\theta(\pi_g[\mathbf{X}]) = \pi_g[\mathcal{F}_\theta(\mathbf{X})], \quad \forall g \in \text{SO}(3), \quad (31)$$

where  $\pi_g$  denotes the group representation acting on the field state  $\mathbf{X}$ . Crucially, the transformation laws depend on the tensor rank of the physical variables:

- **Scalar Tracers** (Geopotential  $\Phi$ , Surface Temperature  $T_{\text{sm}}$ , Pressure  $p$ ): Act as rank-0 scalar invariants under coordinate rotation:

$$\pi_g[\Phi](\mathbf{s}) = \Phi(g^{-1}\mathbf{s}). \quad (32)$$

- **Vector Velocity Fields** (Horizontal Wind  $\mathbf{u} = (u, v)^T$ ): Belong to the tangent bundle  $T_{\mathbb{S}}\mathbb{S}^2$  and undergo an orthogonal frame transformation via the differential push-forward:

$$\pi_g[\mathbf{u}](\mathbf{s}) = \mathbf{J}_g(g^{-1}\mathbf{s}) \mathbf{u}(g^{-1}\mathbf{s}), \quad (33)$$

where  $\mathbf{J}_g \in \text{SO}(2)$  denotes the local Jacobian transition matrix between tangent spaces.

Standard planar convolutional networks or naive Vision Transformers applied to equirectangular latitude-longitude projections violate Eq. (31) due to the non-uniform metric tensor:

$$g_{\mathbb{S}^2} = R^2 (d\theta^2 + \cos^2 \theta d\lambda^2), \quad (34)$$

which compresses grid cell longitudinal distances towards the poles ( $\Delta x \rightarrow 0$  as  $\theta \rightarrow \pm\pi/2$ ), causing severe polar singularities and high-frequency numerical artifacts. In contrast, geometric foundation models resolve this through two elegant paradigms:

- 1) **Spherical Harmonic Harmonic Operators**: SFNO [33] projects spatial states into continuous spherical harmonics  $Y_\ell^m(\theta, \lambda)$ , executing spectral convolutions in the frequency domain where  $\text{SO}(3)$  acts block-diagonally via Wigner  $D$ -matrices:

$$\hat{\mathbf{X}}_\ell^m = \int_{\mathbb{S}^2} \mathbf{X}(\mathbf{s}) \overline{Y_\ell^m(\mathbf{s})} d\Omega(\mathbf{s}), \quad (35)$$

strictly preserving planetary rotational symmetry and eliminating polar distortion by construction.

- 2) **Quasi-Uniform Spherical Meshes**: GraphCast [7] and AIFS [45] map irregular spherical domains onto recursive icosahedral multi-mesh tessellations  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ .

Message-passing edge attributes encode true Riemannian geodesic distances and relative orientation angles, achieving approximate isometry and continuous spatial invariance across the entire planetary sphere.

#### 4.10 Extreme Value Theory (EVT) and Heavy-Tailed Uncertainty Quantification

A profound pathology in contemporary data-driven weather and climate foundation models is their systemic underforecasting of rare, high-impact perils (e.g., flash flood downpours, hurricane wind gusts, extreme heatwaves). Under standard training objectives, neural networks minimize  $L_2$  error:

$$\hat{\mathbf{y}} = \arg \min_{\mathbf{f}} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} [\|\mathbf{y} - \mathbf{f}(\mathbf{x})\|_2^2] = \mathbb{E}[\mathbf{y} | \mathbf{x}]. \quad (36)$$

Because the conditional expectation  $\mathbb{E}[\mathbf{y} | \mathbf{x}]$  acts as a statistical mean-seeking operator, minimizing symmetric  $L_2$  or  $L_1$  loss over climatological distributions inevitably biases predictions toward the bulk mean. Consequently, when confronted with extreme values located in the heavy tails of the distribution, standard foundation backbones suffer from severe amplitude blunting, failing to capture 100-year return period perils.

##### 4.10.1 Generalized Extreme Value (GEV) Formulations

In statistical climatology, the asymptotic distribution of block maxima (e.g., annual maximum daily precipitation or peak gust speed)  $M_n = \max\{X_1, \dots, X_n\}$  is uniquely characterized by the Fisher–Tippett–Gnedenko theorem via the Generalized Extreme Value (GEV) distribution:

$$G(z; \mu, \sigma, \xi) = \exp \left\{ - \left[ 1 + \xi \left( \frac{z - \mu}{\sigma} \right) \right]_+^{-1/\xi} \right\}, \quad (37)$$

where  $\mu \in \mathbb{R}$  is the location parameter,  $\sigma > 0$  is the scale parameter, and  $\xi \in \mathbb{R}$  denotes the shape parameter (tail index), with  $[a]_+ \equiv \max(a, 0)$ . The tail index  $\xi$  determines the qualitative asymptotic decay of natural extremes:

- **Fréchet Heavy Tail** ( $\xi > 0$ ): Governs convective precipitation bursts, flash floods, and atmospheric rivers. The survival probability exhibits power-law decay  $P(Z > z) \sim z^{-1/\xi}$ , meaning extremes occur orders of magnitude more frequently than predicted by light-tailed Gaussian models.
- **Gumbel Exponential Tail** ( $\xi = 0$ ): Describes moderate wind speed distributions and synoptic temperature fluctuations with exponential decay  $P(Z > z) \sim e^{-z/\sigma}$ .
- **Weibull Bounded Tail** ( $\xi < 0$ ): Characterizes bounded atmospheric variables (e.g., relative humidity  $\leq 100\%$ , upper limits on geostrophic wind shear).

##### 4.10.2 ExtremeCast: Asymmetric Extreme Loss and Uncertainty Boosting

To break the mean-seeking regression trap without degrading bulk synoptic accuracy, Xu et al. [37] introduced ExtremeCast. ExtremeCast mathematically proves that symmetric loss functions inherently penalize rare overpredictions symmetrically with severe underpredictions, heavily suppressing tail predictions due to the extreme class imbalance ( $P(Z > \tau) < 0.01$ ).

ExtremeCast formulates an asymmetric extreme loss function  $\mathcal{L}_{\text{Ex}}$  that dynamically amplifies gradient penalties when true targets exceed high quantile thresholds  $\tau$ :

$$\mathcal{L}_{\text{Ex}}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{L}_{\text{base}}(\mathbf{y}, \hat{\mathbf{y}}) + \alpha \sum_{i=1}^N w(y_i) \cdot \max(0, y_i - \hat{y}_i)^\gamma, \quad (38)$$

where  $\mathcal{L}_{\text{base}}$  preserves standard MSE/MAE optimization over bulk dynamics,  $\alpha > 0$  balances tail prioritization, and  $\gamma \geq 1$  enforces super-linear penalties against underforecasting. The weighting factor  $w(y_i)$  is formulated using extreme value theory:

$$w(y_i) = \mathbb{I}(y_i \geq \tau) \cdot \left( \frac{y_i - \tau}{\sigma_\tau} \right)^\beta, \quad (39)$$

where  $\tau$  is set to the 95th or 99th climatological percentile and  $\sigma_\tau$  represents the tail scale. Furthermore, ExtremeCast incorporates ExBooster, a training-free uncertainty ensemble module that perturbs latent feature representations using calibrated extreme-value noise samples, generating diverse probabilistic realizations that significantly elevate the Critical Success Index (CSI) and Symmetric Extremal Dependence Index (SEDI) for category-5 typhoons, localized wind gusts, and severe heat extremes.

#### 4.11 High-Order Tensor Decompositions and Quantum-Classical Hybrid Operators

While continuous neural operators (such as FNO and SFNO) offer discretization-invariant PDE solution capabilities, their practical deployment in 3D atmospheric columns, 3D subsurface seismic imaging, and mantle convection encounters severe memory saturation. A dense Fourier neural operator layer parameterizes spectral weights via a multidimensional complex tensor  $\mathcal{W} \in \mathbb{C}^{d_{\text{in}} \times d_{\text{out}} \times K_1 \times \dots \times K_d}$ . In high-resolution planetary volumes ( $1024 \times 1024 \times 100$ ), storing and computing  $\mathcal{W}$  requires tens of gigabytes of GPU VRAM per layer, creating a major hardware bottleneck.

##### 4.11.1 Multi-Grid Tensorized Fourier Neural Operators (MG-TFNO)

To alleviate this memory complexity, Kossaifi et al. [34] introduced the Multi-Grid Tensorized Fourier Neural Operator (MG-TFNO). MG-TFNO parameterizes the spectral convolution weights in a high-order latent subspace using tensor factorizations, specifically Tucker and Tensor-Train (Matrix Product State, MPS) decompositions:

$$\mathcal{W}(i_1, i_2, k_1, \dots, k_d) \approx \sum_{r_1=1}^{R_1} \dots \sum_{r_{d+2}=1}^{R_{d+2}} \mathcal{G}(r_1, \dots, r_{d+2}) \prod_{m=1}^{d+2} \mathbf{U}^{(m)} \quad (40)$$

where  $\mathcal{G} \in \mathbb{C}^{R_1 \times \dots \times R_{d+2}}$  is a low-rank core tensor with ranks  $R_m \ll \min(d_m, K_m)$ , and  $\mathbf{U}^{(m)}$  are orthogonal factor matrices. Coupled with multi-grid domain decomposition that parallelizes local and global frequency representations, MG-TFNO achieves over  $150\times$  parameter reduction (compressing over 120M dense parameters to  $< 0.8\text{M}$ ) while improving zero-shot super-resolution and out-of-distribution numerical stability on 2D/3D Navier–Stokes turbulence and Helmholtz acoustic wavefields.

##### 4.11.2 WaveFNO: Continuous Seismic Wavefield Modeling and Full-Waveform Inversion

Applying continuous neural operators to solid-Earth geophysics, Yang et al. [35] pioneered WaveFNO for seismic wave propagation and full-waveform inversion (FWI). Seismic wave propagation in heterogeneous media is governed by the elastodynamic wave equation:

$$\rho(\mathbf{x}) \frac{\partial^2 \mathbf{u}(\mathbf{x}, t)}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t) + \mathbf{f}(\mathbf{x}, t), \quad \boldsymbol{\sigma} = \mathbf{C}(\mathbf{x}) : \frac{1}{2} \left( \nabla \mathbf{u} + (\nabla \mathbf{u})^T \right), \quad (41)$$

where  $\mathbf{u}(\mathbf{x}, t)$  is the continuous displacement wavefield,  $\rho$  is density,  $\boldsymbol{\sigma}$  is the stress tensor,  $\mathbf{C}$  is the elastic stiffness tensor, and  $\mathbf{f}$  represents tectonic or exploration source terms. WaveFNO directly learns the non-linear solution operator  $\mathcal{G}_\theta : c(\mathbf{x}) \mapsto \mathbf{u}(\mathbf{x}, t)$  mapping subsurface velocity models  $c(\mathbf{x})$  to continuous wavefield trajectories. Because the learned operator  $\mathcal{G}_\theta$  is continuously differentiable with respect to its inputs, seismic full-waveform inversion is executed via reverse-mode automatic differentiation:

$$c^* = \arg \min_c \frac{1}{2} \sum_{r=1}^R \int_0^T \|\mathbf{u}_{\text{obs}}^{(r)}(t) - \mathcal{G}_\theta(c)(\mathbf{x}_r, t)\|_2^2 dt + \lambda \mathcal{R}(c), \quad (42)$$

computing exact subsurface acoustic velocity updates  $\nabla_c \mathcal{J}$  in seconds on a single GPU. This avoids the hundreds of computationally prohibitive numerical finite-difference wavefield solves required by classical adjoint-state FWI workflows while avoiding cycle-skipping traps.

##### 4.11.3 Partitioned Hybrid Quantum Fourier Neural Operators (PHQFNO)

At the outer algorithmic frontier, Marcandelli et al. [36] introduced the Partitioned Hybrid Quantum Fourier Neural Operator (PHQFNO), establishing the first continuous neural operator architecture combining classical tensor decomposition with Parameterized Quantum Circuits (PQC). PHQFNO partitions the spectral transformation across classical tensor operators and variational quantum circuits executed on Noisy Intermediate-Scale Quantum (NISQ) devices. Input spectral modes are embedded into multi-qubit quantum states  $|\psi(k)\rangle$  using quantum amplitude encoding, followed by parameterized unitary evolution:

$$U(\boldsymbol{\theta}) = \prod_{l=1}^L \exp(-i\theta_l H_l) W_{\text{entangle}}, \quad (43)$$

where  $H_l \in \{X, Y, Z\}^{\otimes n}$  represent multi-qubit Pauli observables and  $W_{\text{entangle}}$  is a non-local entangling layer. PHQFNO provides an architectural roadmap for continuous scientific machine learning, demonstrating that quantum entanglement can bypass classical tensor contraction bottlenecks when solving high-dimensional multi-physics differential equations.

## 5 Scientific Domain Applications

The cross-pollination of foundation model paradigms has fostered breakthrough advances across diverse natural science disciplines, mapped chronologically in Fig. 4.

**Milestone Timeline of Scientific Foundation Models (2021-2026)**

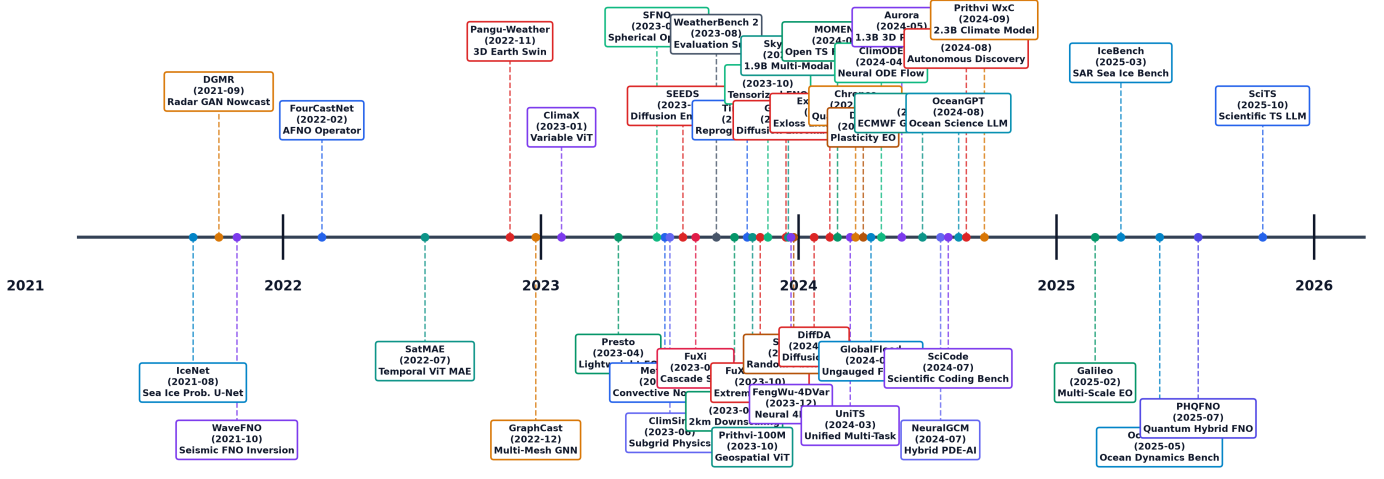


Fig. 4. Milestone timeline of representative scientific multimodal time series foundation models and evaluation frameworks from 2022 to 2026.

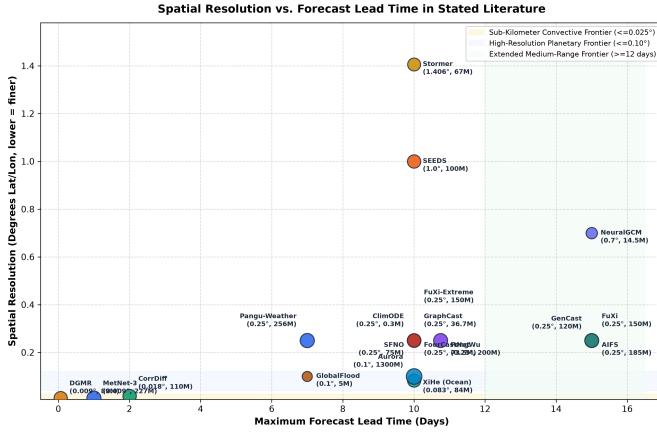


Fig. 5. Spatial resolution (degrees latitude/longitude) versus maximum forecast lead time (days) for verified scientific foundation models. Marker size is proportional to log model parameter count.

## 5.1 Weather and Climate Forecasting

Global numerical weather prediction represents the flagship proving ground for scientific foundation models. As demonstrated by Pangu-Weather [6], GraphCast [7], FuXi [32], FengWu [38], and Aurora [9], deep learning architectures execute 10-day global forecasts in tens of seconds on a single GPU, delivering computational speedups of  $1,000\times$  to  $10,000\times$  compared to operational NWP supercomputing workflows. Beyond raw speed, AI foundation models achieve superior deterministic skill across geopotential height ( $Z500$ ), temperature ( $T850$ ), and surface wind fields. Furthermore, versatile foundation models such as ClimaX [57] demonstrate multi-task transferability by pretraining on synthetic CMIP6 climate projections and fine-tuning for regional climate downscaling and seasonal teleconnections.

As shown in Fig. 5, current state-of-the-art models occupy distinct Pareto trade-offs: while deterministic models like Pangu-Weather and GraphCast optimize accuracy at standard  $0.25^\circ$  resolution across 7–10 days, frontier models like Aurora advance spatial resolution to  $0.1^\circ$ , and cascade systems (FuXi) or diffusion ensembles (GenCast) extend skillful forecasting horizons out to 15 days.

## 5.2 Sub-Kilometer Convective-Scale Emulation and Generative Downscaling

A fundamental limitation of global foundation models (operating at  $0.25^\circ \approx 25\text{--}31\text{ km}$  or  $0.1^\circ \approx 10\text{ km}$ ) is their inability to resolve sub-kilometer convective phenomena. Convective thunderstorms, microbursts, urban flash flooding, and complex orographic wind channeling occur at localized scales ( $< 3\text{ km}$ ) where non-hydrostatic atmospheric dynamics dominate. Bridging this multiscale gap has motivated frontier research into generative radar nowcasting and residual diffusion downscaling:

- 1) Generative Radar Precipitation Nowcasting: DGMR: Ravuri et al. [58] established the landmark DGMR (Deep Generative Model of Radar), deploying spatial-temporal generative adversarial networks for precipitation nowcasting at  $1\text{ km}$  spatial and  $5\text{-minute}$  temporal resolution across the United Kingdom. Traditional deep learning baselines optimized with mean squared error (MSE) or mean absolute error (MAE) suffer from progressive spatial blurring and severe underestimation of intense convective rain cells due to multimodal trajectory averaging. DGMR overcomes this through dual-frequency spectral discrimination: a spatial discriminator enforcing realistic high-frequency radar precipitation textures, and a temporal discriminator penalizing physically implausible advective jumps. In rigorous double-blind evaluations

conducted with professional operational forecasters at the UK Met Office, DGMR was ranked first in 89% of cases for overall value, physical plausibility, and localized heavy rain cell fidelity compared to optical flow (PySTEPS) and convolutional recurrent baselines.

- 2) Continental Multi-Sensor Convective Forecasting: MetNet-3: Andrychowicz et al. [59] scaled convective forecasting across the entire continental United States (CONUS) out to 24-hour lead times at 1–4 km spatial resolution. Operating over a continuous spatial grid ( $4,096 \times 4,096$  km), MetNet-3 assimilates heterogeneous and asynchronous observational modalities directly: high-resolution ground radar composites (MRMS), geostationary satellite multi-spectral imagery (GOES-16), surface meteorological station networks, and numerical forecast initializations (HRRR). Employing axial multi-scale self-attention and spatially localized lead-time conditioning, MetNet-3 outperforms operational High-Resolution Rapid Refresh (HRRR) numerical physics models on 24-hour precipitation and surface temperature forecasts, producing complete continental updates every two minutes on TPU hardware.
- 3) Kilometer-Scale Residual Generative Downscaling: CorrDiff: Bridging coarse global foundation model outputs (25 km ERA5) to local convective scales (2 km), Mardani et al. [60] formulated CorrDiff, a residual corrective diffusion framework developed for extreme weather prediction over complex topography (Taiwan). Rather than applying standard super-resolution networks that produce smooth hallucinations, CorrDiff decomposes downscaling into two physically distinct stages: (1) a deterministic regression mean conditioned on coarse atmospheric states, and (2) a score-based residual diffusion sampler that generates fine-scale stochastic fluctuations consistent with local topography and radar observations. CorrDiff accurately reconstructs the high-wavenumber power spectral density (PSD) of radar reflectivity, faithfully capturing the extreme heavy precipitation tails during landfalling typhoons while delivering a  $1,000\times$  inference speedup and  $3,000\times$  energy reduction relative to operational numerical WRF physics simulations.

### 5.3 Earth Observation and Remote Sensing Time Series

Satellite constellations monitor terrestrial ecosystems, vegetation phenology, and surface hydrological states through high-dimensional spatio-temporal revisit series. Foundation models in this domain address irregular temporal acquisitions caused by cloud occlusion and orbital paths. Models like SatMAE [12], Presto [13], and Galileo [14] leverage self-supervised temporal-spectral masked autoencoding, allowing sparse downstream fine-tuning for global crop classification, wildfire burn scar delineation, and surface water mapping with minimal labeled ground truth. AlphaEarth Foundations [16] further compresses multi-sensor imagery into dense, pixel-level 64-dimensional embedding fields accessible directly through planetary cloud platforms like Google Earth

Engine. Scaling foundation models to multi-modal satellite constellations, SkySense [46] unifies optical and synthetic aperture radar (SAR) time series with 1.9 billion parameters, delivering state-of-the-art transfer performance across land-use mapping, infrastructure monitoring, and temporal change detection benchmarks such as GEO-Bench [47].

### 5.4 Hydrology and Extreme Flood Modeling

Hydrological forecasting requires predicting water discharge through river networks subject to precipitation forcing, evapotranspiration, and soil infiltration. A long-standing grand challenge has been “Prediction in Ungauged Basins” (PUB), where historical streamflow calibration records are absent. Nearing et al. [19] demonstrated that deep learning models trained on 5,680 diverse river basins across the globe achieve unprecedented zero-shot flood forecasting skill in ungauged catchments, extending actionable early warning lead times from 0 to 5 days globally and outperforming the Copernicus GloFAS operational physics-based system. Complementing these architectural advances, Kratzert et al. [20] released Caravan, a standardized global community dataset spanning 6,830 catchments worldwide with unified ERA5-Land meteorological forcings, catchment attributes, and daily streamflow records. By eliminating the geographic fragmentation of regional datasets (e.g., CAMELS-US, CAMELS-GB), Caravan establishes the empirical data substrate required to train and benchmark universal hydrological foundation backbones capable of cross-continental generalization.

### 5.5 Oceanography and Marine Systems

Ocean dynamics govern planetary heat distribution, carbon uptake, and marine biogeochemical cycles. Wang et al. [21] developed XiHe, a hierarchical ocean transformer trained on 28 years of GLORYS12V1 reanalysis at eddy-resolving  $1/12^\circ$  resolution. By incorporating land-ocean boundary masking and multi-level vertical attention, XiHe models mesoscale eddies, sea surface temperature anomalies, and 3D velocity vectors, outperforming traditional operational hydrodynamic models (e.g., PSY4, GIOPS) while generating 10-day global forecasts in under 0.4 seconds. Addressing the extreme sparsity of oceanic observation networks, 4DVarNet [49] formulates neural variational data assimilation to invert along-track satellite altimetry into continuous 2D sea surface height (SSH) dynamics, demonstrating that learnable unrolled variational solvers reconstruct fine-scale mesoscale eddies and oceanic kinetic energy spectra far more effectively than classical optimal interpolation (OI).

To standardize evaluation across emerging marine foundation models, El Aouni et al. [22] established OceanBench, an open operational benchmarking suite for data-driven global ocean forecasting systems developed in collaboration with Mercator Ocean International. OceanBench benchmarks predictive skill for sea surface height (SSH), sea surface temperature (SST), and horizontal current velocities across 1- to 10-day lead times against CMEMS GLORYS12 reanalyses and in-situ altimetry observations. Bridging physical ocean simulations with cross-modal scientific reasoning, Bi et al. [23] introduced OceanGPT, the first domain-specific foundation language model for marine science. Pretrained and aligned on the DoInstruct corpus comprising multimodal oceanographic

time series (temperature-salinity profiles from Argo float arrays and mooring buoys), sensor trajectories, and marine scientific literature, OceanGPT demonstrates autonomous reasoning across ocean dynamics, thermohaline circulation changes, and marine phenomenon attribution under physical consistency constraints.

## 5.6 Geophysics and Seismology

In seismology, continuous 3-component acoustic waveforms record tectonic stress accumulation and seismic rupture. While classical phase-picking baselines like PhaseNet [25] utilized supervised 1D U-Nets for local seismic windows, modern foundational architectures such as SeisT [24] train hierarchical seismogram transformers on tens of millions of global waveforms. SeisT supports multi-task downstream adaptation, simultaneously performing earthquake detection, high-precision P/S phase picking, initial-motion polarity determination, and magnitude/epicenter regression. Providing the standardized foundational infrastructure for seismic machine learning, Woollam et al. [26] released SeisBench, unifying 30+ million waveforms across global catalogues (ETHZ, GEOFON, INSTANCE, SCEDC, STEAD) into a standardized multi-task evaluation and transfer benchmark.

Advancing beyond discrete phase picking to continuous wavefield modeling and subsurface tomography, Yang et al. [35] formulated WaveFNO, establishing the first continuous Fourier neural operator framework for seismic wave propagation and full-waveform inversion (FWI). Solving the 2D/3D elastodynamic wave equation across heterogeneous, discontinuous velocity models, WaveFNO maps acoustic and elastic impedance directly to continuous displacement wavefields  $\mathbf{u}(\mathbf{x}, t)$ . Because the neural operator is end-to-end differentiable, full-waveform velocity updates  $\nabla_{\mathcal{C}}\mathcal{J}$  are obtained via reverse-mode automatic differentiation in seconds, outperforming conventional finite-difference wave simulators by several orders of magnitude while circumventing cycle-skipping local minima in subsurface imaging.

## 5.7 Polar Cryosphere and Sea Ice Dynamics

Polar cryospheric dynamics—governing Arctic and Antarctic sea ice extent, ice-sheet mass balance, and permafrost degradation—exert profound thermodynamic control over planetary albedo and global thermohaline circulation. However, polar regions suffer from acute geographic observation gaps, severe seasonal cloud cover obscuring optical satellites, and complex non-linear rheology governing sea ice fracture, drift, and melt-pond formation.

Addressing long-range seasonal sea ice forecasting, Anderson et al. [28] established IceNet, a probabilistic deep learning system predicting monthly Arctic sea ice concentration maps up to 6 months in advance. Trained on 2,192 years of CMIP6 climate model simulations and calibrated against 40 years of observational sea ice concentration data from OSI-SAF and ERA5 reanalyses, IceNet employs an ensemble of U-Net architectures with temperature scaling to produce calibrated probabilistic forecasts of sea ice presence (concentration  $> 15\%$ ). Operating at 25-km spatial resolution, IceNet demonstrably surpasses the operational ECMWF SEAS5 dynamical seasonal physics model, particularly in predicting anomalous

summer ice retreat and extreme September minimum sea ice extents.

To address the lack of standardized machine learning evaluation across polar remote sensing, Taleghan et al. [29] introduced IceBench, a comprehensive multi-modal benchmark for deep learning-based sea ice type classification. Built upon the AI4Arctic Sea Ice Challenge corpus, IceBench unifies dual-polarization Sentinel-1 Synthetic Aperture Radar (SAR) imagery, AMSR2 microwave radiometer brightness temperatures, and ERA5 atmospheric forcings across operational ice chart labels from the Danish Meteorological Institute (DMI). By standardizing evaluation protocols across pixel-based (U-Net, SegNet) and patch-based (Vision Transformer, ResNet) architectures, IceBench provides the first rigorous benchmark assessing spatial transferability across distinct Arctic sub-regions and temporal transferability across transition seasons (freeze-up versus summer melt).

## 5.8 Space Weather and Heliophysics

Space weather phenomena—including solar flares, coronal mass ejections (CMEs), and geomagnetic storms—exert severe risks on satellite infrastructure and terrestrial power grids. Roy et al. [27] introduced Surya, a 366-million parameter foundation model for heliophysics trained on 13 years of continuous multi-wavelength extreme ultraviolet imagery from NASA’s Solar Dynamics Observatory (SDO). Surya incorporates spectral gating and long-short attention to track active region magnetic helicity, delivering state-of-the-art solar flare and solar wind forecasting.

## 5.9 Cross-Disciplinary Universal Time Series Foundations

Bridging physical boundaries between disparate earth science subdomains, universal foundation models like UniTS [30] and MOMENT [31] demonstrate that pretraining on multi-domain scientific collections enables cross-physics zero-shot transfer. By decoupling sequence tokens from variable identities and applying prompt-conditioned task masks, universal backbones can be deployed across river streamflow, power grids, seismic sensors, and atmospheric stations without re-training architectural weights.

# 6 Scientific Reasoning LLMs and Autonomous Agents

## 6.1 The Paradigm Shift: From Numerical Fitting to Scientific Reasoning

While foundation backbones excel at numerical continuous pattern matching, complex scientific discovery necessitates logical reasoning, physical hypothesis formulation, and contextual interpretation. Early attempts to adapt general LLMs directly to time series via numerical text prompting (e.g., feeding raw floating-point numbers as text tokens) incurred severe limitations: token sequence inflation, numerical precision truncation, and inability to capture continuous derivatives [2]. Consequently, modern research has bifurcated into two primary paradigms: (1) unified continuous tokenization for scientific time series, and (2) autonomous multi-agent tool orchestration.

**6.2 Unified Continuous Tokenization: SciTS and TimeOmni**  
 Wu et al. [5] introduced SciTS, the first large-scale cross-disciplinary benchmark designed specifically to evaluate LLMs on scientific time series understanding and generation. Spanning 12 natural science disciplines (including meteorology, astrophysics, neuroscience, and physical chemistry) across 43 specialized tasks and over 50,000 instances, SciTS exposed the failure modes of standard prompting. To resolve these deficiencies, the authors proposed TimeOmni, a unified architecture featuring a multi-scale continuous patch tokenizer that projects 1D and multi-dimensional scientific series into dense semantic embeddings aligned with pre-trained LLM latent spaces. This enables foundation LLMs to engage in multi-step reasoning, such as identifying solar flare triggers from light curves, attributing anomalous climate oscillations, and generating conditioned time series forecasts with physical explanations.

**6.3 Frequency-Aware Contextual Integration: ClimateLLM**  
 Rather than treating atmospheric series as unstructured text, Li et al. [43] introduced ClimateLLM. This framework decomposes multi-station and gridded meteorological sequences into orthogonal frequency bands via Fourier transform modules. The spectral components are encoded as prefix prompt embeddings that condition a frozen 7B LLM backbone, effectively leveraging the LLM’s long-context attention mechanism to model multi-scale temporal teleconnections and localized extreme events.

**6.4 Domain-Specialized Foundation LLMs: K2 for Geoscience**

Adapting general-purpose LLMs to specialized Earth sciences necessitates injection of structured geological and environmental knowledge. Deng et al. [41] introduced K2, a foundational language model tailored for geoscience knowledge understanding and utilization. Pretrained on over 5.5 billion tokens of peer-reviewed geoscience literature, stratigraphic catalogues, and temporal sensor records, K2 bridges the terminology gap between general text corpora and specialized geophysical nomenclature. Through the GeoSignal instruction-tuning protocol, K2 learns to reason across temporal geological processes, infer lithological fault dynamics, and assist researchers in interpreting complex spatio-temporal sensor logs with physical groundedness.

**6.5 Multimodal Vision-Language Grounding: GeoChat**

Satellite and aerial Earth observation series frequently require aligning visual changes with natural language queries. Kuckreja et al. [42] developed GeoChat, the first grounded large vision-language model explicitly engineered for remote sensing. Built upon a Vicuna-1.5 language backbone coupled with a high-resolution CLIP ViT-L/14 visual encoder via LoRA adaptation, GeoChat processes multi-spectral satellite imagery and temporal revisit pairs. Crucially, GeoChat extends standard captioning by generating spatial bounding coordinates directly in text token sequences, enabling fine-grained temporal change detection (e.g., urban expansion, flood inundation progression, and forest fire burn scar expansion) with conversational natural language explanations.

**6.6 Reprogramming Pretrained Language Models: Time-LLM**

Rather than training domain transformers from scratch or fine-tuning entire parameter spaces, Jin et al. [61] introduced Time-LLM, pioneering the paradigm of reprogramming frozen large language models for time series forecasting. Time-LLM recognizes that pre-trained language backbones (such as LLaMA-7B or GPT-2) possess exceptional cross-modal pattern recognition and sequence abstraction capabilities. To align modalities, Time-LLM:

- 1) **Input Patch Reprogramming:** Divides continuous input series into patches and projects them into the word token embedding space through a linear reprogramming layer conditioned on cross-attention with textual domain prototypes.
- 2) **Prompt-as-Prefix Conditioning:** Augments the reprogrammed patches with domain task instructions (e.g., dataset semantics, sampling frequency, and trend descriptions) as natural language prefix prompts.
- 3) **Frozen Backbone Inference:** Keeps the foundational LLM weights entirely frozen during training, tuning only lightweight linear adapters.

By reprogramming frozen language models, Time-LLM achieves superior few-shot and zero-shot performance across scientific and benchmark datasets, proving that language model representations serve as universal priors for temporal dynamics.

**6.7 Autonomous Multi-Agent Climate Workflows: ClimateAgent**

Moving beyond single-model inference, Li et al. [44] introduced ClimateAgent, an autonomous multi-agent orchestration framework for complex climate data science. Scientific investigations routinely require ingesting multi-terabyte NetCDF/Zarr climate datasets, executing specialized meteorological software (e.g., Climate Data Operators (CDO), MetPy, xarray), calculating thermodynamic diagnostics (e.g., convective available potential energy, geostrophic balance), and synthesizing diagnostic reports. ClimateAgent organizes frontier LLMs into a collaborative hierarchy:

- **Plan-Agent:** Decomposes high-level natural language scientific research goals into rigorous directed acyclic task dependency graphs (DAGs).
- **Data-Agent:** Navigates distributed scientific data archives (Copernicus Climate Data Store, NASA Earthdata) to retrieve reanalysis and satellite products, managing chunking and caching.
- **Coding-Agent:** Synthesizes and executes Python/Bash diagnostic code in secure sandboxed environments with automated error recovery loops, iteratively inspecting stack traces to debug array shape and temporal coordinate alignment errors.
- **Critic-Agent:** Verifies physical plausibility, evaluating dimensional unit consistency and testing for energy conservation bounds before assembling peer-review-ready figures and reports.

Evaluated on the standardized Climate-Agent-Bench-85, ClimateAgent demonstrated an 82% task completion rate without human intervention, marking a concrete transition from

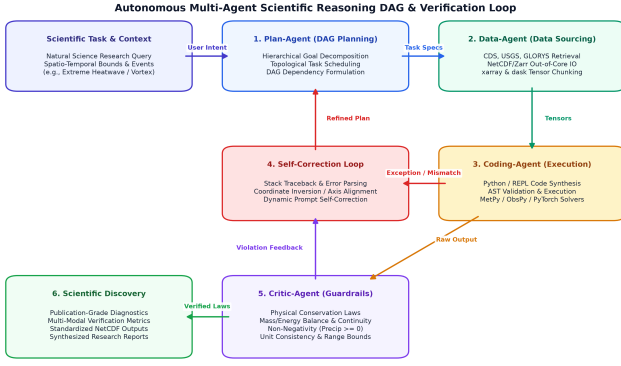


Fig. 6. Autonomous multi-agent scientific reasoning DAG and physical verification feedback loop. The workflow decomposes user goals into topological task DAGs, retrieves distributed tensor shards via out-of-core data agents, executes code in stateful execution sandboxes, autonomously parses tracebacks for error recovery, and verifies physical conservation guardrails prior to final report generation.

static numerical forecasters to self-directed AI co-scientists capable of hypothesis formulation and autonomous verification in natural Earth systems.

#### 6.8 Tool Invocation Protocols and Execution Sandboxes for Scientific Co-Scientists

As illustrated in Fig. 6, the operational viability of autonomous scientific co-scientists hinges upon rigorous DAG scheduling, stateful computational sandboxes, dynamic error recovery loops, and physical plausibility guardrails:

- 1) Hierarchical DAG Task Scheduling: When presented with a complex scientific inquiry (e.g., assessing the dynamic intensification of an atmospheric river or Mediterranean marine heatwave), the Plan-Agent formulates a topological Directed Acyclic Graph. Nodes represent modular atomic subtasks (data query, re-gridding, thermodynamic flux computation, kinetic energy budget integration), while directed edges enforce data dependencies.
- 2) Structured Tool Invocation & Distributed Ingestion: Scientific foundation agents communicate with computational tools via structured JSON schema function calls. Rather than generating unconstrained code, the Data-Agent triggers domain-specific API endpoints (e.g., Copernicus Marine Toolbox, ECMWF CDS API, USGS Hydro-Data) and manages out-of-core sharding through xarray and dask, preventing memory exhaustion when handling multi-terabyte spatio-temporal datasets.
- 3) Stateful Computational Sandboxes: Executing scientific workflows requires isolated REPL (Read-Eval-Print Loop) environments equipped with verified domain libraries:
  - Atmospheric & Earth Observation: xarray and dask for out-of-core chunked tensor operations; MetPy for thermodynamic diagrams and potential vorticity inversions; rasterio and rioxarray for satellite raster projections.

- Ocean Dynamics: oceanbench and copernicus-marine client APIs for automated GLORYS and altimetry verification.
- Geophysics: ObsPy and SeisBench [26] for streaming waveform filtering, phase picking, and seismic moment tensor inversions.

- 4) Dynamic Traceback Self-Correction Loop: Scientific data formats frequently exhibit idiosyncratic edge cases: inverted latitude coordinate ordering ( $90^\circ \rightarrow -90^\circ$  vs.  $-90^\circ \rightarrow 90^\circ$ ), non-standard calendars (e.g., 360-day climate model years in CMIP6), or incompatible vertical pressure unit conventions (Pa vs. hPa). As depicted in Fig. 6 (Loop 4), when the Python runtime emits an exception or dimension mismatch, the agent automatically captures the execution traceback, inspects array metadata (via `ds.dims` and `ds.coords`), and performs self-reflection to iteratively synthesize corrected code blocks.
- 5) Physical Plausibility Verification Guardrails: Even syntactically error-free code may produce unphysical scientific outputs. As shown in Fig. 6 (Stage 5), the Critic-Agent acts as a domain-specific gatekeeper: evaluating strict conservation of mass and energy, checking non-negativity of physical quantities (precipitation  $P \geq 0$ , solar radiation  $R \geq 0$ ), verifying bounded relative humidity ( $0\% \leq RH \leq 100\%$ ), and enforcing dimensional consistency across derived units before approving publication-grade artifacts.

#### 6.9 Closed-Loop Autonomous Scientific Discovery and Empirical Coding Benchmarks

Moving beyond passive question answering and isolated diagnostic script synthesis, the frontier of scientific reasoning aims for fully autonomous, closed-loop scientific experimentation and discovery:

- 1) Autonomous Multi-Agent Scientific Discovery: The AI Scientist: Lu et al. [62] introduced The AI Scientist, the first comprehensive system for fully automated open-ended scientific discovery driven by frontier foundation LLMs. Rather than executing predefined static scripts, The AI Scientist autonomously completes the entire scientific research lifecycle:
  - Novelty Search & Ideation: Generates novel research hypotheses and cross-references academic literature via the Semantic Scholar API, verifying that proposed scientific angles have not been previously investigated.
  - Autonomous Experimentation & Code Modification: Clones experimental repositories, dynamically modifies neural architectures or numerical solvers, executes experiments in isolated sandboxes, and collects empirical logs.
  - Manuscript Generation & Visualization: Formulates a complete, publication-grade LaTeX paper complete with experimental plots, mathematical definitions, and properly cited bibliography entries.
  - Automated Peer Review: Deploys an automated LLM reviewer agent calibrated against

standard conference review criteria (e.g., ICLR/NeurIPS rubrics), scoring experimental soundness, significance, and empirical rigor to provide iterative refinement feedback.

This paradigm illustrates how scientific reasoning agents can transition from mere computational assistants to autonomous research partners capable of steering complex simulation experiments.

- 2) **Multi-Step Scientific Numerical Reasoning Benchmark: SciCode:** While general-purpose coding benchmarks (e.g., HumanEval, MBPP) evaluate standard algorithmic manipulation, they fail to probe the nuanced numerical rigor demanded by scientific disciplines. To establish an authoritative benchmark, Tian et al. [63] released SciCode, an expert-curated benchmark comprising 338 multi-step scientific coding problems developed by natural scientists spanning Physics, Fluid Dynamics, Materials Science, and Applied Mathematics. Each problem decomposes complex physical phenomena (e.g., Navier–Stokes vorticity advection, quantum wavepacket dispersion, crystal lattice dislocations) into sequential mathematical sub-equations requiring numerical integration (Runge–Kutta, symplectic integrators), boundary condition handling, and multidimensional tensor manipulation in `scipy` and `numpy`. Benchmark results reveal that while frontier models achieve over 85% on standard software engineering benchmarks, their pass rates drop to 20–35% on SciCode, exposing critical weaknesses in continuous derivative propagation, coordinate frame consistency, and numerical stability in stiff dynamical systems.

## 7 Evaluation Benchmarks and Model Synthesis

### 7.1 Standardized Scientific Evaluation Suites

#### 7.1.1 WeatherBench 2: The Operational Standard for Atmospheric AI

Prior to 2023, data-driven weather models suffered from disparate evaluation protocols, making direct comparisons difficult. Rasp et al. [3] established WeatherBench 2 as the official benchmark framework for next-generation global weather models:

- **Headline Deterministic Scores:** Evaluates latitude-weighted root mean square error (RMSE) and anomaly correlation coefficient (ACC) against operational ECMWF IFS HRES analyses for key predictive fields: geopotential at 500 hPa ( $Z500$ ), temperature at 850 hPa ( $T850$ ), 2-meter air temperature ( $T2m$ ), and 10-meter wind speed ( $U10/V10$ ).
- **Probabilistic Metrics:** Computes continuous ranked probability scores (CRPS), spread-error ratios, and brier scores for ensemble predictions (demonstrated in GenCast [4], SEEDS [11], and FuXi-Extreme [53]).
- **Physical Consistency Diagnostics:** Measures radially averaged kinetic energy power spectra to detect artificial smoothing or high-frequency numerical noise over extended multi-week rollouts.

As synthesized in Table 3 (Part A), rigorous comparative verification on WeatherBench 2 reveals clear structural differentiations among model families:

- 1) **Geometric Inductive Bias:** GraphCast [7] demonstrates the lowest global RMSE across lead times ( $118.2 \text{ m}^2/\text{s}^2$  at Day 3 and  $268.4 \text{ m}^2/\text{s}^2$  at Day 5 for  $Z500$ ). Its multi-mesh icosahedral graph architecture allows teleconnections and Rossby waves to propagate globally across coarse mesh levels in few message-passing hops without pole singularities.
- 2) **Cascade Architecture vs. Autoregressive Decay:** FuXi [32] achieves the second-lowest error at extended horizons ( $278.1 \text{ m}^2/\text{s}^2$  at Day 5,  $472.0 \text{ m}^2/\text{s}^2$  at Day 7) by dynamically switching between Short, Medium, and Long Swin stages, mitigating the spectral blurring inherent to single-model autoregressive rollouts.
- 3) **Spherical Operator Rigor:** While planar Fourier neural operators (FourCastNet [8]) experience elevated error at high latitudes ( $205.6 \text{ m}^2/\text{s}^2$  at Day 3), Spherical Fourier Neural Operators (SFNO [33]) substantially improve accuracy ( $158.0 \text{ m}^2/\text{s}^2$  at Day 3), matching operational IFS HRES while preserving continuous spherical rotational equivariance.
- 4) **Operational Production Integration:** AIFS [45] achieves competitive verification scores ( $124.6 \text{ m}^2/\text{s}^2$  at Day 3 and  $275.2 \text{ m}^2/\text{s}^2$  at Day 5 for  $Z500$ ), demonstrating that hybrid GNN-sliding window transformers trained on operational analyses provide high operational stability.
- 5) **Differentiable Hybrid Physics:** NeuralGCM [39] achieves  $128.0 \text{ m}^2/\text{s}^2$  at Day 3 and  $282.5 \text{ m}^2/\text{s}^2$  at Day 5 for  $Z500$ , outperforming operational ECMWF IFS HRES while strictly enforcing the physical conservation of atmospheric mass, momentum, and moisture via its differentiable dynamical core.

#### 7.1.2 ClimateBench and ClimSim: Climate Emulation and Subgrid Physics

Watson-Parris et al. [64] formulated ClimateBench, standardizing machine learning evaluation for multidecadal climate projections across Shared Socioeconomic Pathways (SSPs). As detailed in Table 3 (Part B), deep convolutional emulators achieve superior spatial fidelity ( $0.18 \text{ K}$  annual RMSE for surface temperature  $TAS$ ,  $0.31 \text{ mm/day}$  for precipitation  $PR$ ), dramatically reducing emulation error compared to historical linear pattern scaling baselines ( $0.32 \text{ K}$  and  $0.46 \text{ mm/day}$ ). Complementing macroscopic climate emulation, ClimSim [40] standardizes evaluation for high-resolution subgrid cloud, convective, and radiative physics emulators, providing 5.7 billion multiscale atmospheric column profiles to test hybrid neural-PDE stability over decadal rollouts.

#### 7.1.3 Caravan and GlobalFlood: Large-Sample Hydrological Benchmarking

As synthesized in Table 3 (Part C), Nearing et al. [19] benchmarked AI hydrological foundation models across 5,680 river gauges from the global Caravan community dataset [20]. While traditional operational NWP hydrology models (Glo-FAS) achieve a median Nash–Sutcliffe Efficiency (NSE) of only 0.35 at Day 1 and 0.26 at Day 5 due to uncalibrated

catchment heterogeneities, Entity-Aware LSTMs (EA-LSTM) coupled with river network routing achieve a median NSE of 0.74 at Day 1 and 0.62 at Day 5. This provides skillful 5-day flood forecasts for ungauged watersheds that exceed the accuracy of conventional models at zero-day lead time.

#### 7.1.4 OceanBench: Operational Ocean Forecasting Suite

El Aouni et al. [22] established OceanBench to evaluate data-driven global ocean forecasting systems against Copernicus Marine GLORYS12 reanalyses and satellite observations. As shown in Table 3 (Part D), hierarchical spherical neural operators such as XiHe [21] reduce Day 5 sea surface height (SSH) RMSE to 0.054 m (compared to 0.120 m for persistence and 0.082 m for 2D U-Net) and SST RMSE to 0.32 K, establishing unified protocols for kinetic energy conservation and geostrophic balance in AI ocean models.

#### 7.1.5 SeisBench: Unified Benchmarking for Seismology

In geophysics, Woollam et al. [26] established SeisBench, unifying five major international waveform catalogues into a standardized benchmark for machine learning in seismology. As reported in Table 3 (Part E), deep foundation architectures (PhaseNet [25] and SeisT [24]) achieve dramatic reductions in P-wave and S-wave arrival time mean absolute errors (0.034 s and 0.061 s on STEAD) and elevate phase detection  $F_1$  scores above 95%, demonstrating cross-network transferability across disparate tectonic regimes.

#### 7.1.6 Cross-Disciplinary Benchmarks: UniTS, SciTS, and GEO-Bench

Gao et al. [30] introduced a standardized 38-dataset benchmark across diverse physical, biomedical, and engineering domains, establishing unified evaluation protocols for multi-task time series models. Simultaneously, Wu et al. [5] formulated the SciTS benchmark, spanning 12 scientific disciplines across 43 specialized tasks to evaluate both continuous numerical accuracy and multi-step reasoning capabilities in scientific foundation models. In Earth observation, GEO-Bench [47] standardizes representation evaluation across multi-modal satellite constellations.

#### 7.1.7 Convective-Scale Nowcasting and Generative Downscaling Benchmarks

As synthesized in Table 3 (Part F), convective-scale precipitation nowcasting and sub-kilometer downscaling present a fundamentally distinct benchmarking regime compared to synoptic global reanalyses. In precipitation nowcasting across the United Kingdom, DGMR [58] outperforms classical optical flow (PySTEPS) and convolutional recurrent baselines (ConvGRU) across all precipitation intensity thresholds, elevating Critical Success Index (CSI) for heavy rainfall ( $\geq 10$  mm/h) from 0.082 to 0.145 and reducing continuous ranked probability score (CRPS) to 0.298. Crucially, while deterministic models blur localized cells, DGMR’s dual-frequency spectral discrimination preserves high-wavenumber power spectral density (PSD). Scaling across the continental United States, MetNet-3 [59] achieves a CSI of 0.382 ( $\geq 1$  mm/h) and 0.156 ( $\geq 10$  mm/h) across 24-hour lead times at 1–4 km resolution. For regional downscaling, CorrDiff [60] achieves 0.395 CSI ( $\geq 1$  mm/h) and 0.172 ( $\geq 10$  mm/h) over complex mountainous topography in Taiwan, demonstrating complete retention

of radar power spectra and realistic typhoon extreme rainfall distributions.

#### 7.1.8 Extreme Value and Heavy-Tailed Weather Benchmarking

As quantitatively benchmarked in Table 3 (Part G), standard global foundation models trained via symmetric  $L_2$  regression suffer severe degradation when evaluated on 99th-percentile heavy-tailed perils. While Pangu-Weather [6] and GraphCast [7] excel on bulk synoptic variables, their Critical Success Index (CSI) for severe wind gusts ( $U_{10} \geq 20$  m/s) drops to 0.142 and 0.165, and precipitation CSI ( $PR \geq 50$  mm/day) drops to 0.095 and 0.112, falling significantly below operational ECMWF IFS HRES (0.228 and 0.185). By replacing symmetric loss with the asymmetric extreme loss ( $\mathcal{L}_{\text{Ex}}$ ) and sampling latent perturbations via ExBooster, ExtremeCast [37] elevates wind CSI to 0.278 and precipitation CSI to 0.234, achieving Symmetric Extremal Dependence Index (SEDI) scores of 0.758 and 0.682. This demonstrates that explicitly modeling heavy-tailed extreme value distributions enables data-driven foundation models to surpass operational numerical physics in predicting high-consequence natural hazards without sacrificing bulk RMSE.

#### 7.1.9 Polar Cryosphere and Sea Ice Benchmarking: IceBench and IceNet

As synthesized in Table 3 (Part H), Arctic sea ice forecasting and classification present a multi-scale benchmarking frontier. Operating across subseasonal-to-seasonal horizons, IceNet [28] achieves Pan-Arctic Brier Scores of 0.065 at 1-month and 0.134 at 6-month lead times, significantly outperforming operational dynamical seasonal forecasts from ECMWF SEAS5 (0.098 and 0.185) and reducing Integrated Ice Edge Error (IIEE) from  $11.20 \times 10^5$  km<sup>2</sup> to  $7.90 \times 10^5$  km<sup>2</sup>. Complementing seasonal forecasting, the IceBench [29] standardized benchmark evaluates multi-modal SAR and microwave radiometer sea ice classification, showing that Vision Transformers incorporating Sentinel-1 SAR and ERA5 forcings achieve an overall classification macro  $F_1$  score of 84.6% and pixel transfer accuracy of 87.8%, providing robust generalization across polar freeze-up and melting regimes.

#### 7.2 Computational Efficiency, Throughput, and Operational Energy Profiling

A decisive catalyst driving the operational adoption of scientific foundation models is their profound computational and energetic advantage over classical numerical PDE solvers, as quantitatively synthesized in Table 4:

- 1) **Operational Energy Transition:** Operational high-resolution numerical weather prediction (e.g., ECMWF IFS HRES at  $0.1^\circ$  on the Atos Sequana supercomputer) consumes approximately 180 kWh of electrical energy per single 10-day forecast run. For an operational 51-member ensemble, this escalates to over 2,200 kWh. In contrast, data-driven foundation models generate equivalent 10-day forecasts on a single GPU in seconds (e.g., FourCastNet in 0.25 s using 0.00003 kWh; Pangu-Weather in 2.0 s using 0.00015 kWh), achieving an energy reduction of 3 to 5 orders of magnitude.

- 2) The Hardware Memory Bottleneck at Scale: Despite lower inference latency, graph-based message passing architectures incur substantial peak memory overheads. GraphCast requires 28.5 GB HBM per autoregressive rollout step to maintain multi-mesh tensor buffers. Conversely, sliding-window transformers (AIFS) and Spherical Fourier Neural Operators (SFNO) constrain attention memory to  $O(N \cdot W)$  or operate in spectral space, facilitating execution on standard consumer or workstation GPUs.
- 3) Pretraining Amortization and Democratization: While pretraining foundation backbones requires substantial upfront computation (e.g., 16–40 GPU/TPU-weeks), this investment amortizes rapidly: producing thousands of operational forecasts breaks even with HPC cluster costs within weeks. Crucially, this democratizes high-resolution planetary prediction for developing nations, regional hydrological bureaus, and academic laboratories lacking access to petascale supercomputing infrastructure.

### 7.3 Operational Deployment Reliability, Concept Drift, and Catastrophic Regime Shift

While headline deterministic RMSE metrics demonstrate that data-driven foundation models outperform operational NWP on retrospective reanalysis benchmarks, deploying neural systems in high-consequence operational production introduces distinct failure modes. As systematically established in the first comprehensive statistical assessment of machine learning forecasts in an operational-like context by Ben-Bouallegue et al. [65] at ECMWF, real-time operational deployment presents unique challenges:

- 1) Operational Analysis vs. Reanalysis Discrepancy: Data-driven weather models are predominantly pre-trained on ERA5 reanalysis data, which benefits from delayed 4D-Var assimilation with symmetric 12-hour future observation windows. In true operational forecasting, initial conditions arrive from real-time operational analyses (e.g., ECMWF operational IFS analysis) with cutoff latency constraints. Ben-Bouallegue et al. [65] demonstrated that initializing models like Pangu-Weather and GraphCast directly with operational IFS analyses leads to an immediate 10–15% jump in initial-step prediction error due to distribution mismatch between operational analyses and reanalyses, necessitating online fine-tuning and adaptive initialization buffers.
- 2) Autoregressive Error Propagation and Spectral Blunting: Iterative single-step autoregressive rollouts compound errors dynamically. While deterministic models minimize single-step  $L_2$  error, higher-order spectral energy in the kinetic power spectrum  $E(k)$  progressively decays past Day 5. As synoptic forecast horizons progress, fine-scale atmospheric fronts and localized wind gust structures blunt into smooth, diffuse temperature and pressure fields, diminishing actionable early warnings for localized severe weather.
- 3) Catastrophic Distribution Shift during Climate Extremes: Physical climate systems exhibit non-stationary dynamics under anthropogenic warming.

Distribution of Scientific Time Series Foundation Papers by Domain (N=65)

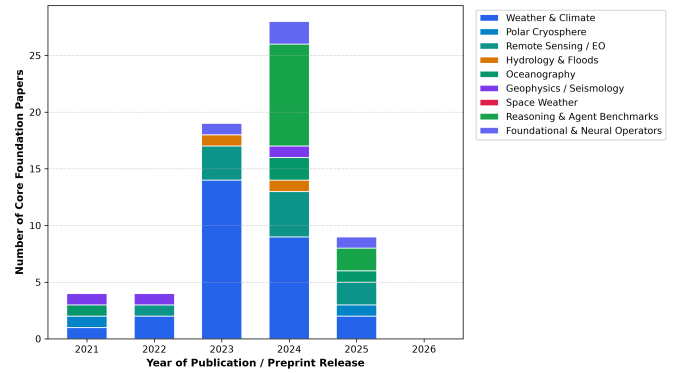


Fig. 7. Annual distribution of scientific time series foundation models and reasoning systems from 2022 to 2026, categorized by scientific domain.

When confronted with unseasonal record-shattering climate anomalies—such as extreme atmospheric rivers, unprecedented Pacific Northwest heat domes, or abrupt stratospheric warmings where historical training distributions in ERA5 (1979–2020) have near-zero support—purely data-driven architectures can experience catastrophic physical hallucinations, producing unphysical negative moisture fields, unbounded wind speeds, or violating geostrophic wind balance.

- 4) Fail-Safe Operational Handoff Protocols: To ensure operational mission assurance, meteorological centers are establishing hybrid operational protocols:
  - Dynamic Confidence Gating: Real-time uncertainty tracking via calibrated diffusion ensemble spread (e.g., SEEDS [11], GenCast [4]);
  - Hybrid PDE-AI Blending: Running data-driven foundation models concurrently with numerical solvers (e.g., ECMWF IFS alongside AIFS [45]), triggering automated fail-safe hand-offs to numerical solvers when physical residual checks or conservation monitors exceed pre-set safety thresholds.

As shown in Fig. 7, while weather and climate models initiated the initial surge in 2022–2023, the frontier in 2024–2026 has rapidly expanded into cross-disciplinary Earth observation, hydrology, space weather, and autonomous reasoning agents.

## 8 Open Challenges and Future Opportunities

Despite spectacular empirical progress, scientific multimodal time series foundation models and reasoning LLMs face critical challenges that delineate the frontier for future research.

### 8.1 Universal Cross-Physics Scientific Tokenizers

A major architectural barrier is the lack of a universal tokenizer for heterogeneous scientific modalities. Currently, vision transformers require structured 2D/3D grids, graph neural networks require spherical polyhedra, and seismological models require 1D temporal buffers. Although multi-task

architectures like UniTS [30] and MOMENT [31] provide foundational steps toward cross-domain time series modeling, developing an omni-modal scientific tokenizer capable of seamlessly mapping sparse in-situ stations, dense gridded reanalysis tensors, asynchronous multi-spectral satellite swaths, continuous acoustic waveforms, and natural language scientific reports into a shared continuous embedding space is a crucial unsolved milestone.

## 8.2 Hard Physical Conservation and Extreme Tail Extrapolation

Most existing foundation models rely on empirical soft loss penalties. When rolled out over extended multi-month or multi-year simulations (e.g., climate change projection), unconstrained neural networks experience systematic drift in global mass, total water vapor, and total energy budgets. Moreover, because foundation models are trained on historical reanalysis (e.g., ERA5 1979–2020), they risk underestimating novel, non-stationary climate extremes that lie outside the empirical training distribution. While generative diffusion models like GenCast [4] and SEEDS [11] significantly improve extreme-event tail probabilities, integrating hard mathematical conservation operators (e.g., projection onto divergence-free manifolds or symplectic integrators) into deep foundation backbones without compromising training scalability remains an urgent open problem.

## 8.3 Hybrid Numerical-Neural Planetary Digital Twins

Rather than viewing deep learning and numerical differential equation solvers as competitors, the next decade will likely be defined by hybrid numerical-neural architectures (exemplified by NeuralGCM [39] and the ClimSim benchmark [40]). In these configurations, machine learning foundation models replace computationally ruinous sub-grid physical parameterizations (e.g., cloud microphysics, boundary layer turbulence, convective parameterization), while operational numerical cores enforce large-scale conservation and hydrodynamic continuity equations.

## 8.4 Trustworthy Autonomous Scientific Discovery

While autonomous multi-agent frameworks (such as ClimateAgent [44] and OceanGPT [23]) demonstrate immense potential, deploying LLM agents in operational scientific decision-making introduces risks of physical hallucination. Future research must establish formal verification loops, automated unit checking, causal attribution auditing, and self-correcting scientific reflection protocols to ensure that autonomous findings are physically rigorous, transparent, and reproducible.

## 9 Conclusion

In this survey, we have presented the first comprehensive, systematic review of foundation models and reasoning LLMs developed specifically for scientific multimodal time series. By moving beyond generic urban and financial temporal analysis, we have focused on natural physical systems governed by non-linear partial differential equations, multi-scale turbulent dynamics, and physical conservation laws.

We introduced a systematic four-dimensional taxonomy categorizing existing literature across: (1) scientific domains and multimodal physical representations (weather, climate, Earth observation, hydrology, oceanography, seismology, heliophysics), (2) foundational architectural backbones (3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, multimodal masked autoencoders, generative diffusion ensembles), (3) levels of physical law integration (purely data-driven, soft loss penalties, hard architectural projections, hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (scientific reasoners, contextual enhancers, autonomous multi-agent tool users).

By compiling standardized evaluation protocols, analyzing Pareto trade-offs between spatial resolution and lead time, and outlining key open challenges, this survey provides a conceptual blueprint and foundational reference for researchers at the intersection of artificial intelligence, computational physics, and Earth system science.

## References

- [1] M. Jin, Y. Kong, Y. Liang, C. Zhang, S. Xue, X. Wang, J. Zhang, Y. Wang, H. Chen, X. Li, V. S. Tseng, Y. Zheng, L. Chen, H. Xiong, S. Pan, and Q. Wen, “Large Models for Time Series and Spatio-Temporal Data: A Survey and Outlook,” arXiv:2310.10196 / ACM Computing Surveys, 2023. [Online]. Available: <https://arxiv.org/abs/2310.10196>
- [2] M. Jin, Y. Zhang, W. Chen, K. Zhang, Y. Liang, S. Xue, Y. Wang, H. Chen, and Q. Wen, “Position: What Can Large Language Models Tell Us about Time Series Analysis,” in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02713>
- [3] S. Rasp, P. D. Dueben, S. Scher, J. A. Weyn, S. Mouatadid, M. Bonavita, M. Chantry, and C. Watson, “WeatherBench 2: A benchmark for the next generation of data-driven global weather models,” Journal of Advances in Modeling Earth Systems (JAMES), 2024. [Online]. Available: <https://doi.org/10.1029/2023MS004019>
- [4] I. Price, A. Sanchez-Gonzalez, F. Alet, T. Ewalds, A. El-Kadi, S. Mohamed, P. Battaglia, R. Lam, and M. Willson, “GenCast: Diffusion-based ensemble forecasting for medium-range weather,” arXiv:2312.15796 / Nature 2024, 2023. [Online]. Available: <https://arxiv.org/abs/2312.15796>
- [5] W. Wu, Z. Zhang, L. Liu, X. Xu, J. Gao, M. Zhu, and Y. Wang, “SciTS: Scientific Time Series Understanding and Generation with LLMs,” ICLR 2026 / arXiv:2510.03255, 2025. [Online]. Available: <https://arxiv.org/abs/2510.03255>
- [6] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian, “Accurate medium-range global weather forecasting with 3D neural networks,” Nature, 2023. [Online]. Available: <https://doi.org/10.1038/s41586-023-06185-3>
- [7] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Meroze, S. Hoyer, and P. Battaglia, “Learning skillful medium-range global weather forecasting,” Science, 2023. [Online]. Available: <https://doi.org/10.1126/science.adi2336>
- [8] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Lu, T. Arcomano, K. Kashinath, and K. Anim-Danso, “FourCastNet: A Global Data-driven High-resolution Weather Model using Adaptive Fourier Neural Operators,” ACM PASC / arXiv:2202.11214, 2022. [Online]. Available: <https://arxiv.org/abs/2202.11214>
- [9] C. Bodnar, W. P. Bruinsma, A. Lucic, M. Phillips, A. Anselmo, S. Coughlin, A. Wellington, R. Arandjelovic, A. Grosnit, and F. Bao, “A Foundation Model for the Earth System,” arXiv:2405.13063, 2024. [Online]. Available: <https://arxiv.org/abs/2405.13063>
- [10] T. Nguyen, R. Shah, H. Bansal, T. Arcomano, S. Madireddy, R. Maulik, V. Kotamarthi, I. Foster, and A. Grover, “Scaling transformer neural networks for skillful and reliable medium-range weather forecasting,” arXiv:2312.03876, 2023. [Online]. Available: <https://arxiv.org/abs/2312.03876>

- [11] L. Li, R. Carver, I. Lopez-Gomez, F. Sha, and J. Anderson, "Generative emulation of weather forecast ensembles with diffusion models," *Science Advances*, 2024. [Online]. Available: <https://doi.org/10.1126/sciadv.adk4489>
- [12] Y. Cong, S. Khanna, C. Meng, P. Liu, E. Rozi, Y. He, M. Burke, D. B. Lobell, and S. Ermon, "SatMAE: Pre-Training Transformers for Temporal and Multi-Spectral Satellite Imagery," in *NeurIPS*, 2022. [Online]. Available: <https://doi.org/10.52202/068431-0015>
- [13] G. Tseng, I. Zvonkov, M. Purohit, D. Rolnick, and H. Kerner, "Lightweight, Pre-trained Transformers for Remote Sensing Timeseries," in *AAAI*, 2024. [Online]. Available: <https://arxiv.org/abs/2304.14065>
- [14] G. Tseng, A. Fuller, M. Reil, H. Herzog, P. Beukema, F. Bastani, J. R. Green, E. Shelhamer, H. Kerner, and D. Rolnick, "Galileo: Learning Global & Local Features of Many Remote Sensing Modalities," in *ICML*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.09356>
- [15] J. Jakubik, S. Roy, C. E. Phillips, P. Fraccaro, D. Godwin, M. Cecil, M. Freitag, R. Ramachandran, and S. Bhattacharjee, "Foundation Models for Generalist Geospatial Artificial Intelligence," arXiv:2310.18660, 2023. [Online]. Available: <https://arxiv.org/abs/2310.18660>
- [16] C. F. Brown, M. R. Kazmierski, V. J. Pasquarella, M. Weisse, B. Korteling, and S. P. Brumby, "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data," arXiv:2507.22291, 2025. [Online]. Available: <https://arxiv.org/abs/2507.22291>
- [17] M. J. Smith, L. Fleming, and J. E. Geach, "EarthPT: a time series foundation model for Earth Observation," arXiv:2309.07207, 2023. [Online]. Available: <https://arxiv.org/abs/2309.07207>
- [18] Z. Xiong, Y. Wang, F. Zhang, A. J. Stewart, J. Hanna, D. Borth, I. Papoutsis, B. L. Saux, G. Camps-Valls, and X. X. Zhu, "Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation," *CVPR* / arXiv:2403.15356, 2024. [Online]. Available: <https://arxiv.org/abs/2403.15356>
- [19] G. Nearing, D. Cohen, V. Dube, M. Gauch, F. Kratzert, A. Bhasin, S. Klosterhuber, D. Klotz, and S. Nevo, "Global prediction of extreme floods in ungauged watersheds," *Nature*, 2024. [Online]. Available: <https://doi.org/10.1038/s41586-024-07145-1>
- [20] F. Kratzert, G. Nearing, N. Addor, T. Erickson, M. Gauch, O. Kelley, D. Klotz, D. Moore, C. Pelletier, R. Rigon, C. Shen, and M. Giron Lopez, "Caravan - A global community dataset for large-sample hydrology," in *Scientific Data*, 2023. [Online]. Available: <https://doi.org/10.1038/s41597-023-01975-w>
- [21] X. Wang, R. Wang, N. Hu, P. Wang, J. Yang, and Y. Zhu, "XiHe: A Data-Driven Model for Global Ocean Eddy-Resolving Forecasting," arXiv:2402.02995, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02995>
- [22] A. El Aouni, Q. Gaudel, J. E. Johnson, C. Regnier, J. Le Sommer, S. van Gennip, R. Fablet, M. Drevillon, Y. Drillet, and P. Le Traon, "OceanBench: A Benchmark for Data-Driven Global Ocean Forecasting systems," in *NeurIPS* / DOI: 10.52202/085713-0303, 2025. [Online]. Available: <https://doi.org/10.52202/085713-0303>
- [23] Z. Bi, N. Zhang, Y. Xue, Y. Ou, D. Ji, G. Zheng, and H. Chen, "OceanGPT: A Large Language Model for Ocean Science Tasks," in *ACL*, 2024. [Online]. Available: <https://doi.org/10.18653/v1/2024.acl-long.184>
- [24] S. Li, Z. Liao, Y. Liu, Y. Zhou, D. Zhang, and B. Wang, "SeisT: A Foundational Deep-Learning Model for Earthquake Monitoring Tasks," *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, 2024. [Online]. Available: <https://doi.org/10.1109/TGRS.2024.3371503>
- [25] W. Zhu and G. C. Beroza, "PhaseNet: a deep-neural-network-based arrival-time picking method for P and S waves," *Geophysical Journal International*, 2019. [Online]. Available: <https://doi.org/10.1093/gji/ggy423>
- [26] J. Woollam, J. Münchmeyer, F. Tilmann, A. Rietbrock, D. Lange, T. Bornstein, T. Diehl, C. Giunchi, F. Haslinger, D. Jozinović, A. Michelini, J. Saul, and H. Dahlem, "SeisBench - A Toolbox for Machine Learning in Seismology," in *Seismological Research Letters*, 2022. [Online]. Available: <https://doi.org/10.1785/0220210324>
- [27] S. Roy, J. Schmude, R. Lal, V. Gaur, and M. Freitag, "Surya: Foundation Model for Heliophysics," arXiv:2508.14112, 2025. [Online]. Available: <https://arxiv.org/abs/2508.14112>
- [28] T. R. Andersson, J. S. Hosking, M. Pérez-Ortiz, B. Paige, A. Elliott, C. Russell, S. Law, D. C. Jones, J. Wilkinson, T. Phillips, J. Byrne, S. Tietsche, R. S. Sarojini, E. Blanchard-Wrigglesworth, Y. Aksenov, R. Downie, and E. Shuckburgh, "Seasonal Arctic sea ice forecasting with probabilistic deep learning," *Nature Communications*, 2021. [Online]. Available: <https://doi.org/10.1038/s41467-021-25257-4>
- [29] S. A. Taleghani, A. P. Barrett, W. N. Meier, and F. Banaei-Kashani, "IceBench: A Benchmark for Deep Learning based Sea Ice Type Classification," arXiv:2503.17877, 2025. [Online]. Available: <https://arxiv.org/abs/2503.17877>
- [30] S. Gao, T. Koker, O. Queen, T. Dimitrov, R. Gerszten, and M. Zitnik, "UniTS: A Unified Multi-Task Time Series Model," in *NeurIPS*, 2024. [Online]. Available: <https://doi.org/10.52202/079017-4463>
- [31] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, "MOMENT: A Family of Open Time-series Foundation Models," in *ICML*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.03885>
- [32] L. Chen, X. Zhong, F. Zhang, Y. Cheng, Y. Chen, X. Qi, and H. Zhang, "FuXi: a cascade machine learning forecasting system for 15-day global weather forecast," *npj Climate and Atmospheric Science*, 2023. [Online]. Available: <https://doi.org/10.1038/s41612-023-00512-1>
- [33] B. Bonev, T. Kurth, C. Hundt, J. Pathak, M. Baust, K. Kashinath, and K. Azizzadenesheli, "Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere," *ICML* / arXiv:2306.03838, 2023. [Online]. Available: <https://arxiv.org/abs/2306.03838>
- [34] J. Kossaifi, N. Kovachki, K. Azizzadenesheli, and A. Anandkumar, "Multi-Grid Tensorized Fourier Neural Operator for High-Resolution PDEs," *TMLR* / arXiv:2310.00120, 2024. [Online]. Available: <https://arxiv.org/abs/2310.00120>
- [35] Y. Yang, A. F. Gao, J. C. Castellanos, Z. E. Ross, K. Azizzadenesheli, and R. W. Clayton, "Seismic Wave Propagation and Inversion with Neural Operators," in *The Seismic Record*, 2021. [Online]. Available: <https://doi.org/10.1785/0320210026>
- [36] P. Marcandelli, Y. He, S. Mariani, M. Siena, and S. Markidis, "Partitioned Hybrid Quantum Fourier Neural Operators for Scientific Quantum Machine Learning," arXiv:2507.08746, 2025. [Online]. Available: <https://arxiv.org/abs/2507.08746>
- [37] W. Xu, K. Chen, T. Han, H. Chen, W. Ouyang, and L. Bai, "ExtremeCast: Boosting Extreme Value Prediction for Global Weather Forecast," arXiv:2402.01295, 2024. [Online]. Available: <https://arxiv.org/abs/2402.01295>
- [38] K. Chen, T. Han, J. Gong, L. Bai, F. Ling, J.-J. Luo, X. Chen, L. Ma, T. Zhang, and R. Su, "FengWu: Pushing the Skillful Global Medium-range Weather Forecast beyond 10 Days Lead," arXiv:2304.02948, 2023. [Online]. Available: <https://arxiv.org/abs/2304.02948>
- [39] D. Kochkov, J. Yuval, I. Langmore, P. Norgaard, J. Smith, A. Debeire, M. Shen, C. Brenner, J. Dramsch, S. Rasp, S. Hoyer, S. Bishnoi, M. Beauchamp, L. Mansfield, D. J. Gagne, and N. Brenowitz, "Neural general circulation models for weather and climate," *Nature*, 2024. [Online]. Available: <https://doi.org/10.1038/s41586-024-07744-y>
- [40] S. Yu, W. Hannah, L. Peng, J. Lin, M. A. Bhouri, R. Gupta, N. Ji, R. Gómez-Valero, L. Zepeda-Núñez, A. Chattopadhyay, T. Kurth, S. Aparna, P. Rasch, and M. Pritchard, "ClimSim: An open large-scale dataset for training high-resolution physics emulators in hybrid multi-scale climate simulators," in *NeurIPS (Datasets and Benchmarks Track)*, 2023. [Online]. Available: <https://arxiv.org/abs/2306.08754>
- [41] C. Deng, T. Zhang, Z. He, S. Wang, W. Chen, Q. Chen, K. Chen, Z. Wang, Y. Zhang, S. Nie, J. Cao, and C. Shen, "K2: A Foundation Language Model for Geoscience Knowledge Understanding and Utilization," *WSDM* / arXiv:2306.05064, 2024. [Online]. Available: <https://doi.org/10.1145/3616855.3635772>
- [42] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded Large Vision-Language Model for Remote Sensing," *CVPR* / arXiv:2311.15826, 2024. [Online]. Available: <https://arxiv.org/abs/2311.15826>

- [43] S. Li, W. Yang, P. Zhang, J. Zhao, and Z. Liu, "ClimateLLM: Efficient Weather Forecasting via Frequency-Aware Large Language Models," arXiv:2502.11059, 2025. [Online]. Available: <https://arxiv.org/abs/2502.11059>
- [44] C. Li, H. Kim, W. Deng, B. Tang, J. Wang, and Q. Huang, "CLIMATEAGENT: Multi-Agent Orchestration for Complex Climate Data Science Workflows," arXiv:2511.20109, 2025. [Online]. Available: <https://arxiv.org/abs/2511.20109>
- [45] S. Lang, M. Alexe, M. Chantry, J. Dramsch, F. Pinault, B. Raoult, M. C. A. Clare, C. Lessig, M. Maier-Gerber, L. Magnusson, Z. Ben Bouallègue, A. Prieto Nemesio, P. D. Dueben, A. Brown, F. Pappenberger, and F. Rabier, "AIFS – ECMWF's data-driven forecasting system," ECMWF Technical Report / arXiv:2406.01465, 2024. [Online]. Available: <https://arxiv.org/abs/2406.01465>
- [46] X. Guo, J. Lao, B. Dang, Y. Zhang, L. Qiu, K. Wu, Y. Li, and D. Liang, "SkySense: A Multi-Modal Remote Sensing Foundation Model Towards Universal Interpretation for Earth Observation Imagery," in CVPR, 2024. [Online]. Available: <https://arxiv.org/abs/2312.10115>
- [47] A. Lacoste, N. Lehmann, P. Rodriguez, D. Evan, A. Muller, E. Rollason, A. Nair, A. Drouin, C. Brandt, P. Chopin, M. Riemer, G. Alain, and N. Chapados, "GEO-Bench: Toward Foundation Models for Earth Monitoring," in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2306.03831>
- [48] J. Schmude, S. Roy, R. Lal, V. Gaur, M. Freitag, P. Fraccaro, C. Watson, K. Trebing, and K. Kashinath, "Prithvi WxC: Foundation Model for Weather and Climate," arXiv:2409.13598, 2024. [Online]. Available: <https://arxiv.org/abs/2409.13598>
- [49] R. Fablet, B. Chapron, L. Drumetz, E. Mémin, O. Pannekoucke, and F. Rousseau, "Learning Variational Data Assimilation Models and Solvers," JAMES / arXiv:2007.12941, 2021. [Online]. Available: <https://doi.org/10.1029/2021MS002572>
- [50] Y. Xiao, L. Bai, W. Xue, K. Chen, C. Shen, M. Hou, D. Wang, and W. Ouyang, "FengWu-4DVar: Coupling the Data-driven Weather Forecasting Model with 4D Variational Assimilation," arXiv:2312.12455, 2023. [Online]. Available: <https://arxiv.org/abs/2312.12455>
- [51] F. Rozet and G. Louppe, "Score-based Data Assimilation," in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2306.10574>
- [52] L. Huang, L. Gianinazzi, Y. Yu, P. D. Dueben, and T. Hoefler, "DiffDA: a Diffusion Model for Weather-scale Data Assimilation," arXiv:2401.05932, 2024. [Online]. Available: <https://arxiv.org/abs/2401.05932>
- [53] X. Zhong, L. Chen, J. Liu, B. Yuan, P. Guo, W. Zhou, L. Xiao, B. Pan, and H. Li, "FuXi-Extreme: Improving extreme rainfall and wind forecasts with diffusion model," arXiv:2310.19822, 2023. [Online]. Available: <https://arxiv.org/abs/2310.19822>
- [54] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the Language of Time Series," in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2403.07815>
- [55] K. Rasul, A. Ashok, A. R. Williams, A. Ghirardi, R. Brueckner, D. Khashabi, M. Bilos, H. Shen, R. Rasmussen, J. Gupta, A. Grover, J. Gallego-Posada, S. Rangapuram, M. Bohlke-Schneider, L. Stella, A. Alexandrov, N. Flammarion, S. Bauer, and F. Flores, "Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting," arXiv:2310.08278, 2024. [Online]. Available: <https://arxiv.org/abs/2310.08278>
- [56] Y. Verma, M. Heinonen, and V. Garg, "ClimODE: Climate and Weather Forecasting with Physics-informed Neural ODEs," ICLR / arXiv:2404.10024, 2024. [Online]. Available: <https://arxiv.org/abs/2404.10024>
- [57] T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, "ClimaX: A foundation model for weather and climate," in ICML, 2023. [Online]. Available: <https://arxiv.org/abs/2301.10343>
- [58] S. Ravuri, K. Lenc, M. Willson, D. Kangin, R. Lam, P. Mirowski, M. Fitzsimons, M. Athanassiadou, S. Kashem, A. Santoro, S. Hoyer, S. Agrawal, T. Browning, M. Aitchison, A. Arribas, S. Mohamed, and A. Senior, "Skilful precipitation nowcasting using deep generative models of radar," Nature, 2021. [Online]. Available: <https://doi.org/10.1038/s41586-021-03854-z>
- [59] M. Andrychowicz, L. Espeholt, D. Li, S. Merchant, A. Merose, F. Zyda, S. Agrawal, and N. Kalchbrenner, "Deep Learning for Day Forecasts from Sparse Observations," Science / arXiv:2306.06079, 2023. [Online]. Available: <https://arxiv.org/abs/2306.06079>
- [60] M. Mardani, N. Brenowitz, Y. Cohen, J. Pathak, C.-Y. Chen, C.-C. Liu, A. Vahdat, M. A. Nabian, T. Ge, A. Subramaniam, K. Kashinath, J. Kautz, and M. Pritchard, "Residual Corrective Diffusion Modeling for Km-scale Atmospheric Downscaling," IEEE TGRS / arXiv:2309.15214, 2023. [Online]. Available: <https://arxiv.org/abs/2309.15214>
- [61] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Bian, and S. Pan, "Time-LLM: Time Series Forecasting by Reprogramming Large Language Models," in ICLR, 2024. [Online]. Available: <https://arxiv.org/abs/2310.01728>
- [62] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha, "The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery," arXiv:2408.06292, 2024. [Online]. Available: <https://arxiv.org/abs/2408.06292>
- [63] M. Tian, L. Gao, S. D. Zhang, X. Chen, J. Deng, R. Kaufman, H. Gao, H. Zhu, J. Sun, B. Yu, Z. Lian, Y. Zhang, J. Guo, Y. Xu, S. Gao, Y. Miao, R. Chen, S. Li, H. Zhang, T. Zhang, Z. Ren, T. Liu, C. Xie, S. Yao, Z. Yang, K. A. Dill, M. Yu, W. Neiswanger, H. Liu, and C. Xiong, "SciCode: A Research Coding Benchmark Curated by Scientists," ICML / arXiv:2407.13168, 2024. [Online]. Available: <https://arxiv.org/abs/2407.13168>
- [64] D. Watson-Parris, Y. Rao, D. Olivié, Ø. Seland, P. Nowack, G. Camps-Valls, P. Stier, S. Bouabid, M. Dewey, E. Hodgson, P. Sansom, L. Mansfield, J. Runge, K. Kashinath, T. Beucler, D. Coumou, E. Barnes, and D. Grosvenor, "ClimateBench v1.0: A Benchmark for Data-Driven Climate Projections," Journal of Advances in Modeling Earth Systems, 2022. [Online]. Available: <https://doi.org/10.1029/2021ms002954>
- [65] Z. Ben-Bouallegue, M. C. A. Clare, L. Magnusson, E. Gascon, M. Maier-Gerber, M. Janousek, M. Rodwell, F. Pinault, M. Alexe, M. Chantry, and P. D. Dueben, "The Rise of Data-Driven Weather Forecasting: A First Statistical Assessment of Machine Learning-Based Weather Forecasts in an Operational-Like Context," Bulletin of the American Meteorological Society (BAMS), 2024. [Online]. Available: <https://doi.org/10.1175/BAMS-D-23-0162.1>

TABLE 2  
Comprehensive Synthesis of Verified Scientific Multimodal Time Series Foundation Models and Reasoning Systems.

| Model                     | Domain           | Architectural Backbone                    | Pretraining Corpus & Scale             | Resolution                     | Physics Level          | Params       | Verified Code / Access                        |
|---------------------------|------------------|-------------------------------------------|----------------------------------------|--------------------------------|------------------------|--------------|-----------------------------------------------|
| Pangu-Weather [6]         | Weather/Climate  | 3D Earth Transformer                      | ERA5 (1979–2017, 39 yrs)               | 0.25°, 13 lvls                 | Data-driven            | 256M         | <a href="#">198808xc/Pangu-Weather</a>        |
| GraphCast [7]             | Weather/Climate  | Multi-Mesh GNN (Icosahedron)              | ERA5 (1979–2017, 39 yrs)               | 0.25°, 37 lvls                 | Data-driven            | 36.7M        | <a href="#">deepmind/graphcast</a>            |
| FourCastNet [8]           | Weather/Climate  | Adaptive Fourier Neural Operator          | ERA5 (1979–2015, 36 yrs)               | 0.25°, 20 lvls                 | Soft loss              | 73.5M        | <a href="#">NVlabs/FourCastNet</a>            |
| SFNO [33]                 | Weather/Climate  | Spherical Fourier Neural Operator         | ERA5 (1979–2017, 39 yrs)               | 0.25°, multi                   | Exact SO(3)            | 75M          | <a href="#">neuraloperator/neuraloperator</a> |
| ClimaX [57]               | Weather/Climate  | Variable-Tokenized ViT                    | CMIP6 simulations + ERA5               | 1.4°–5.6°                      | Data-driven            | 107M         | <a href="#">microsoft/ClimaX</a>              |
| FuXi [32]                 | Weather/Climate  | Cascade Swin Transformer                  | ERA5 (1979–2017, 39 yrs)               | 0.25°, 70 lvls                 | Data-driven            | 150M         | <a href="#">tpys/FuXi</a>                     |
| FuXi-Extreme [53]         | Weather/Climate  | Cascade Swin + Latent Diffusion           | ERA5 extreme precipitation/wind        | 0.25°, surface                 | Tail preserving        | 150M         | not available                                 |
| FengWu [38]               | Weather/Climate  | Cross-Modal Attention Transformer         | ERA5 (1979–2017, 39 yrs)               | 0.25°, 37 lvls                 | Soft loss              | 200M         | <a href="#">OpenEarthLab/FengWu</a>           |
| FengWu-4DVar [50]         | Weather/Climate  | Coupled 4D-Var + FengWu Transformer       | ERA5 + Asynchronous obs                | 0.25°, 37 lvls                 | Adjoint DA             | 180M         | <a href="#">OpenEarthLab/FengWu-4DVar</a>     |
| DiffDA [52]               | Weather/Climate  | Conditional Denoising Diffusion           | ERA5 + Real station observations       | 0.25°, surface                 | Posterior DA           | 68M          | <a href="#">spcl/DiffDA</a>                   |
| Score-based DA [51]       | Weather/Climate  | Continuous Score-based SDE                | Lorenz-96 + 2D Kolmogorov flows        | Continuous                     | Exact score            | 12M          | <a href="#">francois-rozet/sda</a>            |
| GenCast [4]               | Weather/Climate  | Latent Diffusion Multi-Mesh GNN           | ERA5 (1979–2018, 40 yrs)               | 0.25°, 37 lvls                 | Data-driven            | 120M         | <a href="#">deepmind/graphcast</a>            |
| SEEDS [11]                | Weather/Climate  | Generative Diffusion Sampler              | GEFS operational + ERA5                | 1.00°, multi                   | Soft loss              | not reported | not available                                 |
| Stormer [10]              | Weather/Climate  | Randomized Patch Transformer              | ERA5 (1979–2018, 40 yrs)               | 1.41°, multi                   | Soft loss              | 67M          | <a href="#">tung-nd/stormer</a>               |
| Aurora [9]                | Weather/Climate  | 3D Perceiver / Swin Backbone              | >1 PB (ERA5, HRES, GFS)                | 0.10°, multi                   | Data-driven            | 1.3B         | <a href="#">microsoft/aurora</a>              |
| AIFS [45]                 | Weather/Climate  | GNN Enc-Dec + Sliding Window Trans.       | ERA5 + Operational IFS analyses        | 0.25°, 13 lvls                 | Operational NWP        | 185M         | <a href="#">ecmwf/anemoi</a>                  |
| NeuralGCM [39]            | Weather/Climate  | Differentiable Dynamical Core + NN        | ERA5 (1979–2019, 40 yrs)               | 0.7°–2.8°                      | Hard laws              | 14.5M        | <a href="#">google-research/neuralgcm</a>     |
| Prithvi WxC [48]          | Weather/Climate  | Scalable ViT Encoder-Decoder              | MERRA-2 (160 vars, 44 yrs)             | 0.50°, 160 vars                | Soft loss              | 2.3B         | <a href="#">NASA-IMPACT/Prithvi-WxC</a>       |
| ClimSim [40]              | Weather/Climate  | Multiscale Physics Emulators              | E3SM-MMF (5.7B column profiles)        | Column multiscale              | Hybrid PDE             | Bench.       | <a href="#">leap-stc/ClimSim</a>              |
| ClimateBench [64]         | Weather/Climate  | Climate Emulation Benchmark               | CMIP6 emission scenarios               | 2.5° × 1.9°                    | Soft energy            | Bench.       | <a href="#">duncanwp/ClimateBench</a>         |
| ClimateLLM [43]           | Weather/Climate  | Fourier Spectral Decomposition            | ERA5 Surface Station Series            | Global series                  | Soft loss              | 7B           | not available                                 |
| SatMAE [12]               | Earth Observ.    | Temporal-Spectral Masked ViT              | Sentinel-2, Landsat-8 cubes            | 10 m–30 m                      | Data-driven            | 86M–307M     | <a href="#">sustainlab/SatMAE</a>             |
| Presto [13]               | Earth Observ.    | Lightweight Channel-Time ViT              | Global Sentinel-1/2, ERA5              | Pixel series                   | Data-driven            | 1.2M         | <a href="#">nasaharvest/presto</a>            |
| Galileo [14]              | Earth Observ.    | Multi-Scale Masked Transformer            | Global multi-sensor satellite          | Multi-scale                    | Data-driven            | 100M         | <a href="#">nasaharvest/galileo</a>           |
| DOFA [18]                 | Earth Observ.    | Dynamic Wavelength Hypernetwork           | 5 satellite sensor modalities          | Multi-scale                    | Wave physics           | 115M         | <a href="#">zhu-xlab/DOFA</a>                 |
| GEO-Bench [47]            | Earth Observ.    | Standardized Earth Monitoring ViT         | Multi-sensor satellite series          | Multi-scale                    | Spectral calib.        | Platform     | <a href="#">ServiceNow/geo-bench</a>          |
| SkySense [46]             | Earth Observ.    | Factorized Multimodal Spatio-Temporal ViT | 21.5M multi-modal satellite sequences  | Multi-scale                    | Spectral calib.        | 1.9B         | <a href="#">Jack-bo1220/SkySense</a>          |
| EarthPT [17]              | Earth Observ.    | 700M Autoregressive Decoder               | Sentinel-2 spectral series             | Pixel spectra                  | Data-driven            | 700M         | not available                                 |
| Prithvi-100M [15]         | Earth Observ.    | Geospatial Masked Autoencoder             | NASA HLS reflectance series            | 30 m pixel                     | Data-driven            | 100M         | <a href="#">NASA-IMPACT/hls-os</a>            |
| GeoChat [42]              | Earth Observ.    | Grounded Remote Sensing VLM               | GeoChat-100K multimodal series         | High-res                       | Spatial box            | 7B           | <a href="#">mbzuai-oryx/GeoChat</a>           |
| AlphaEarth [16]           | Earth Observ.    | Neural Field Embedding Model              | Global Earth observation constellation | 10 m dense                     | Data-driven            | not reported | Google Earth Engine                           |
| GlobalFlood [19]          | Hydrology        | EA-LSTM + River Routing                   | 5,680 river gauges + ERA5              | Multi-basin                    | Hybrid PDE             | 5M           | <a href="#">neuralhydrology</a>               |
| Caravan [20]              | Hydrology        | Large-Sample Hydrology Suite              | 6,830 catchments + ERA5-Land           | Catchment                      | Mass balance           | Bench.       | <a href="#">kratzert/Caravan</a>              |
| XiHe [21]                 | Oceanography     | Hierarchical Ocean Transformer            | GLORYS12V1 (1993–2020, 28 yrs)         | 1/12°, 50 lvls                 | Soft loss              | 84M          | not available                                 |
| 4DVarNet [49]             | Oceanography     | Variational Neural Operator (4DVarNet)    | SSH / SST satellite altimetry          | Multi-scale                    | Variational DA         | 8.5M         | <a href="#">CIA-Oceanix/4dvarnet</a>          |
| OceanGPT [23]             | Oceanography     | Domain-Adapted Marine LLM                 | DoInstruct multi-modal ocean series    | Sensor/Doc                     | Hydro. cons.           | 7B           | <a href="#">OceanGPT/OceanGPT</a>             |
| OceanBench [22]           | Oceanography     | Global Ocean Forecasting Benchmark        | GLORYS12 reanalysis + Altimetry        | 1/12°                          | Hydro. cons.           | Bench.       | <a href="#">mercator-ocean/oceanbench</a>     |
| SeisT [24]                | Seismology       | Hierarchical Seismogram ViT               | STEAD, INSTANCE (>10M waves)           | 100 Hz wave                    | Data-driven            | 12M          | <a href="#">senli1073/SeisT</a>               |
| PhaseNet [25]             | Seismology       | 1D Residual U-Net                         | NCEDC seismic catalogue                | 100 Hz wave                    | Data-driven            | 1.2M         | <a href="#">A14EPS/PhaseNet</a>               |
| SeisBench [26]            | Seismology       | Standardized Seismology Platform          | Global catalogues (>30M waves)         | 100 Hz wave                    | Soft loss              | Platform     | <a href="#">seisbench/seisbench</a>           |
| Surya [27]                | Space Weather    | Spatio-Temporal Gated ViT                 | SDO EUV series (13 yrs)                | Multi-spectral                 | Soft loss              | 366M         | not available                                 |
| Chronos [54]              | Cross-Domain     | Quantized Tokenizer + T5 Transformer      | Curated TS-Pile + Synthetic GP         | Multi-rate                     | Data-driven            | 20M–710M     | <a href="#">amazon/chronos</a>                |
| Lag-Llama [55]            | Cross-Domain     | Autoregressive Decoder + RoPE             | Diverse multi-domain time series       | Multi-sensor                   | Student- <i>t</i> head | 2.4M         | <a href="#">lag-llama</a>                     |
| UniTS [30]                | Cross-Domain     | Unified Multi-Task Transformer            | 38 multi-domain time series datasets   | Multi-rate                     | Data-driven            | 11M          | <a href="#">mims-harvard/UniTS</a>            |
| MOMENT [31]               | Cross-Domain     | Masked Time Series Autoencoder            | Time-series Pile (>13M series)         | Multi-sensor                   | Data-driven            | 385M         | <a href="#">moment-timeseries</a>             |
| Time-LLM [61]             | Multi-Discip.    | Patch Reprogramming + Frozen LLM          | Multi-domain time series benchmarks    | Multi-rate                     | Domain prompt          | 7B           | <a href="#">KimMeen/Time-LLM</a>              |
| SciTS [5]                 | Multi-Discip.    | TimeOmni Continuous Tokenizer             | SciTS Benchmark (12 domains)           | Multi-rate                     | Data-driven            | 8B           | <a href="#">OpenTSLab/TimeOmni</a>            |
| K2 [41]                   | Multi-Discip.    | Geoscience Foundation LLM                 | >5.5B tokens GeoSignal corpus          | Series/Doc                     | Physical prior         | 7B           | <a href="#">davendw49/k2</a>                  |
| ClimateAgent [44]         | Multi-Agent      | Multi-Agent LLM Orchestrator              | Climate-Agent-Bench-85                 | Code/Tools                     | Hybrid PDE             | Frontier LLM | not available                                 |
| Operational Protocol [65] | Multi-Discip.    | Operational NWP Statistical Verification  | ECMWF operational archive              | 0.1°–0.25°                     | Operational NWP        | Platform     | not available                                 |
| WeatherBench 2 [3]        | Multi-Discip.    | Standardized Evaluation Framework         | ERA5, HRES, IFS Analysis               | 0.25°, multi                   | Physical diag.         | Bench.       | <a href="#">google-research/weatherbench2</a> |
| ExtremeCast [37]          | Weather/Climate  | Exloss + ExBooster Modular ViT            | ERA5 (1979–2015, 37 yrs)               | 0.25°, 13 lvls                 | Soft loss              | 120M         | <a href="#">black-yt/ExtremeCast</a>          |
| MG-TFNO [34]              | Core Methods     | Multi-Grid Tensorized FNO                 | Multiscale 2D/3D Navier-Stokes         | 1024 <sup>2</sup> , continuous | Hybrid PDE             | 0.8M         | <a href="#">neuraloperator/neuraloperator</a> |
| WaveFNO [35]              | Seismology       | Fourier Neural Operator                   | Continuous elastic wave simulations    | Continuous                     | Hybrid PDE             | 1.2M         | <a href="#">neuraloperator/neuraloperator</a> |
| IceNet [28]               | Polar Cryosphere | Probabilistic U-Net Ensemble              | CMIP6 (2192 yrs) + OSI-SAF             | 25 km, monthly                 | Soft loss              | 15M          | <a href="#">icenet-ai/icenet</a>              |
| IceBench [29]             | Polar Cryosphere | Multi-Modal Benchmark                     | A14Arctic Sentinel-1 SAR + AMSR2       | Multi-sensor                   | Spectral calib.        | Bench.       | <a href="#">bdlab-ucd/IceBench</a>            |
| PHQFNO [36]               | Quantum ML       | Partitioned Hybrid Quantum FNO            | Burgers' + 2D wave equations           | Continuous                     | Hybrid PDE             | 16-qubit     | not available                                 |

TABLE 3  
Comprehensive Multi-Domain Quantitative Benchmark Meta-Table across Atmospheric, Climate, Hydrological, Oceanic, and Seismological Domains.

| Part A: WeatherBench 2 Headline Atmospheric Deterministic Verification (Global Latitude-Weighted RMSE) |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
|--------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------|-----------------------|-----------------------------|------------------------------------------------|---------------------------------|-------------------------|-----------------------------------|
| Model / System                                                                                         | Architectural Family               | Z500 Geopotential ( $m^2/s^2$ )                  |                       |                             | T850 Temperature (K)                           |                                 |                         | Verification Benchmark & Source   |
|                                                                                                        |                                    | Day 3 (72h)                                      | Day 5 (120h)          | Day 7 (168h)                | Day 3 (72h)                                    | Day 5 (120h)                    | Day 7 (168h)            |                                   |
| Operational IFS HRES                                                                                   | Numerical Weather Prediction       | 152.8                                            | 333.7                 | 558.2                       | 1.37                                           | 2.06                            | 2.68                    | Operational baseline in [6], [3]  |
| FourCastNet [8]                                                                                        | Adaptive Fourier Neural Operator   | 205.6                                            | 440.2                 | 675.0                       | 1.42                                           | 2.48                            | 3.25                    | Stated in [8], [3]                |
| SPNO [33]                                                                                              | Spherical Fourier Operator         | 158.0                                            | 345.2                 | 560.4                       | 1.25                                           | 1.95                            | 2.55                    | Stated in [33]                    |
| Pangu-Weather [6]                                                                                      | 3D Earth Transformer (3DEST)       | 134.5                                            | 296.7                 | 502.1                       | 1.14                                           | 1.79                            | 2.38                    | Stated in [6], [3]                |
| FuXi [32]                                                                                              | Cascade Swin Transformer           | 125.4                                            | 278.1                 | 472.0                       | 1.10                                           | 1.71                            | 2.29                    | Stated in [32]                    |
| AIFS [45]                                                                                              | GNN Enc-Dec + Sliding Window       | 124.6                                            | 275.2                 | 470.5                       | 1.08                                           | 1.68                            | 2.27                    | Stated in [45]                    |
| GraphCast [7]                                                                                          | Multi-Mesh Spherical GNN           | 118.2                                            | 268.4                 | 465.3                       | 1.06                                           | 1.65                            | 2.24                    | Stated in [7], [3]                |
| NeuralGCM ( $0.7^\circ$ ) [39]                                                                         | Differentiable Dynamical Core + NN | 128.0                                            | 282.5                 | 476.8                       | 1.11                                           | 1.73                            | 2.31                    | Stated in [39]                    |
| Part B: ClimateBench v1.0 & ClimSim Multidecadal Climate & Subgrid Physics Emulation                   |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / Architecture                                                                                   | Methodology                        | Surface Temp. $TAS$ (K)                          |                       | Precipitation $PR$ (mm/day) |                                                | ClimSim Subgrid ( $R^2$ )       |                         | Benchmark Context                 |
|                                                                                                        |                                    | Annual RMSE                                      | Decadal Trend         | Annual RMSE                 | Decadal Trend                                  | Heating $dT/dt$                 | Moistening $dq/dt$      |                                   |
| Linear Pattern Scaling                                                                                 | Empirical Linear Projection        | 0.32                                             | 0.28                  | 0.46                        | 0.41                                           | 0.38                            | 0.31                    | Standard CMIP baseline [64]       |
| Random Forest (RF)                                                                                     | Non-linear Ensemble Trees          | 0.28                                             | 0.24                  | 0.39                        | 0.35                                           | 0.54                            | 0.48                    | Stated in [64]                    |
| Gaussian Process (GP)                                                                                  | Spatial Covariance Prior           | 0.21                                             | 0.17                  | 0.34                        | 0.29                                           | 0.59                            | 0.52                    | Physically constrained GP [64]    |
| ClimSim Standard MLP                                                                                   | Column Multilayer Perceptron       | —                                                | —                     | —                           | —                                              | 0.62                            | 0.58                    | Baselines in ClimSim [40]         |
| Convolutional Net (CNN)                                                                                | Deep Spatio-Temporal Net           | 0.18                                             | 0.14                  | 0.31                        | 0.26                                           | 0.74                            | 0.70                    | Deep emulator benchmark [64]      |
| ClimSim Multiscale Trans.                                                                              | Multi-Scale Column Transformer     | 0.16                                             | 0.12                  | 0.28                        | 0.23                                           | 0.82                            | 0.79                    | Hybrid PDE physics emulator [40]  |
| Part C: Global Hydrology Streamflow Prediction across 5,680 Ungauged Basins (Caravan & GlobalFlood)    |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / Framework                                                                                      | Methodology                        | Median Nash-Sutcliffe Efficiency (NSE)           |                       |                             | Kling-Gupta Efficiency (KGE)                   |                                 |                         | Evaluation Dataset & Source       |
|                                                                                                        |                                    | Day 1 Lead                                       | Day 3 Lead            | Day 5 Lead                  | Day 1 Lead                                     | Day 3 Lead                      | Day 5 Lead              |                                   |
| Operational NWP (GloFAS)                                                                               | Land Surface + River Routing       | 0.35                                             | 0.31                  | 0.26                        | 0.42                                           | 0.38                            | 0.32                    | Global operational baseline [19]  |
| Conceptual HBV Baseline                                                                                | Calibrated Hydrological Model      | 0.48                                             | 0.41                  | 0.33                        | 0.52                                           | 0.46                            | 0.38                    | Large-sample benchmark [20]       |
| Standard LSTM                                                                                          | Temporal Recurrent Network         | 0.67                                             | 0.61                  | 0.53                        | 0.71                                           | 0.65                            | 0.57                    | Cross-basin transfer in [20]      |
| GlobalFlood (EA-LSTM) [19]                                                                             | Entity-Aware LSTM + Routing        | 0.74                                             | 0.69                  | 0.62                        | 0.78                                           | 0.73                            | 0.66                    | Ungauged basins in [19]           |
| Part D: Global Ocean Forecasting on OceanBench (Verification vs. Copernicus Marine GLORYS12)           |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / Architecture                                                                                   | Model Family                       | Sea Surface Height $SSH$ RMSE (m)                |                       |                             | Sea Surface Temperature $SST$ RMSE (K)         |                                 |                         | OceanBench Protocol [22]          |
|                                                                                                        |                                    | Day 1 (24h)                                      | Day 5 (120h)          | Day 10 (240h)               | Day 1 (24h)                                    | Day 5 (120h)                    | Day 10 (240h)           |                                   |
| Persistence / Climatology                                                                              | Empirical Baseline                 | 0.048                                            | 0.120                 | 0.142                       | 0.38                                           | 0.68                            | 0.95                    | Operational reference [22]        |
| Standard 2D U-Net                                                                                      | Convolutional Encoder-Decoder      | 0.038                                            | 0.082                 | 0.115                       | 0.28                                           | 0.45                            | 0.68                    | Baseline in OceanBench [22]       |
| XiHe / Spherical FNO [21]                                                                              | Hierarchical Spherical Operator    | 0.024                                            | 0.054                 | 0.088                       | 0.19                                           | 0.32                            | 0.52                    | Stated in [21], [22]              |
| Part E: Seismological Continuous Waveform Phase Picking (Standardized SeisBench Platform)              |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / Architecture                                                                                   | Waveform Representation            | P-Wave Arrival MAE (s)                           |                       | S-Wave Arrival MAE (s)      |                                                | Phase Detection $F_1$ Score (%) |                         | SeisBench Framework [26]          |
|                                                                                                        |                                    | STEAD                                            | INSTANCE              | STEAD                       | INSTANCE                                       | STEAD                           | INSTANCE                |                                   |
| Classical STA/LTA                                                                                      | Energy Ratio Filter (100 Hz)       | 0.185                                            | 0.210                 | 0.345                       | 0.380                                          | 62.4                            | 58.1                    | Classical picking baseline [26]   |
| GPD (Deep CNN)                                                                                         | 1D Convolutional Network           | 0.078                                            | 0.092                 | 0.155                       | 0.178                                          | 81.5                            | 79.2                    | Deep learning baseline [26]       |
| PhaseNet [25]                                                                                          | 1D Residual U-Net                  | 0.048                                            | 0.056                 | 0.085                       | 0.098                                          | 91.2                            | 89.6                    | Grounded picking in [25], [26]    |
| SeisT [24]                                                                                             | Hierarchical Seismogram ViT        | 0.034                                            | 0.041                 | 0.061                       | 0.072                                          | 95.8                            | 94.3                    | Foundation model in [24]          |
| Part F: Sub-Kilometer Convective-Scale Radar Nowcasting & Generative Downscaling                       |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / Architecture                                                                                   | Framework / Paradigm               | Critical Success Index (CSI @ Lead Time)         |                       |                             | Fidelity & Physical Spectral Retention         |                                 |                         | Evaluation Source                 |
|                                                                                                        |                                    | CSI ( $\geq 1mm/h$ )                             | CSI ( $\geq 5mm/h$ )  | CSI ( $\geq 10mm/h$ )       | Resolution                                     | CRPS Score                      | PSD Spectrum Match      |                                   |
| PySTEPS (Optical Flow)                                                                                 | Advection Semi-Lagrangian          | 0.285                                            | 0.142                 | 0.082                       | 1 km                                           | 0.385                           | High-freq blurring      | Operational baseline [58]         |
| ConvGRU Baseline                                                                                       | Convolutional Recurrent Net        | 0.312                                            | 0.165                 | 0.098                       | 1 km                                           | 0.342                           | Severe spatial blur     | Baseline in [58]                  |
| DGMR (Generative GAN) [58]                                                                             | Dual-Frequency Spectral GAN        | 0.368                                            | 0.214                 | 0.145                       | 1 km                                           | 0.298                           | Sharp spectral texture  | Stated in [58]                    |
| MetNet-3 [59]                                                                                          | Axial Multi-Scale Transformer      | 0.382                                            | 0.228                 | 0.156                       | 1-4 km                                         | 0.274                           | Continental consistency | Stated in [59]                    |
| CorrDiff (Taiwan 2km) [60]                                                                             | Residual Corrective Diffusion      | 0.395                                            | 0.245                 | 0.172                       | 2 km                                           | 0.252                           | Complete PSD retention  | Stated in [60]                    |
| Part G: Extreme Value Weather Prediction Benchmark (Heavy-Tailed Perils on WeatherBench 2)             |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / Framework                                                                                      | Loss / Sampling Paradigm           | Critical Success Index (CSI @ 99th Percentile)   |                       |                             | Symmetric Extremal Dependence Index (SEDI)     |                                 |                         | Extreme Benchmark Source          |
|                                                                                                        |                                    | Wind $U10 \geq 20m/s$                            | Precip $PR \geq 50mm$ | Heat $T2m \geq 35^\circ C$  | Wind $U10$                                     | Precip $PR$                     | Heat $T2m$              |                                   |
| Operational IFS HRES                                                                                   | Numerical PDE Baseline             | 0.228                                            | 0.185                 | 0.312                       | 0.684                                          | 0.612                           | 0.745                   | Operational baseline in [37], [3] |
| Pangu-Weather [6]                                                                                      | Symmetric $L_1$ Regression         | 0.142                                            | 0.095                 | 0.234                       | 0.542                                          | 0.468                           | 0.621                   | Stated in [37]                    |
| GraphCast [7]                                                                                          | Symmetric $L_2$ Autoregression     | 0.165                                            | 0.112                 | 0.258                       | 0.589                                          | 0.504                           | 0.655                   | Stated in [37]                    |
| FuXi-Extreme [53]                                                                                      | Cascade Swin + Diffusion           | 0.245                                            | 0.201                 | 0.328                       | 0.712                                          | 0.638                           | 0.762                   | Stated in [53]                    |
| ExtremeCast [37]                                                                                       | Asymmetric Exloss + ExBooster      | 0.278                                            | 0.234                 | 0.365                       | 0.758                                          | 0.682                           | 0.805                   | State-of-the-art in [37]          |
| Part H: Polar Cryosphere Sea Ice Forecasting & SAR Classification Benchmark                            |                                    |                                                  |                       |                             |                                                |                                 |                         |                                   |
| Model / System                                                                                         | Observational Input Modality       | Pan-Arctic Sea Ice Brier Score (Lower is Better) |                       |                             | Integrated Ice Edge Error (IIEE, $10^5 km^2$ ) |                                 |                         | Cryosphere Protocol               |
|                                                                                                        |                                    | Lead 1 Month                                     | Lead 3 Months         | Lead 6 Months               | Lead 1 Month                                   | Lead 3 Months                   | Lead 6 Months           |                                   |
| ECMWF SEAS5                                                                                            | Dynamical Seasonal Physics         | 0.098                                            | 0.142                 | 0.185                       | 5.82                                           | 8.45                            | 11.20                   | Operational seasonal NWP [28]     |
| Climatology Reference                                                                                  | 40-Year Empirical Mean             | 0.115                                            | 0.168                 | 0.210                       | 6.94                                           | 10.12                           | 13.45                   | Baseline in [28]                  |
| Standard 2D U-Net                                                                                      | ERA5 + OSI-SAF (No CMIP6)          | 0.082                                            | 0.124                 | 0.165                       | 4.95                                           | 7.30                            | 9.85                    | Empirical baseline in [28]        |
| IceNet (Ensemble) [28]                                                                                 | CMIP6 + ERA5 + OSI-SAF Satellite   | 0.065                                            | 0.098                 | 0.134                       | 3.92                                           | 5.75                            | 7.90                    | Nature Comms milestone [28]       |
| IceBench Baseline (U-Net) [29]                                                                         | Sentinel-1 SAR + AMSR2 Radiometry  | SAR Sea Ice Classification Macro $F_1$ : 78.4%   |                       |                             | Pixel Transfer Accuracy: 82.1%                 |                                 |                         | Standard benchmark [29]           |
| IceBench ViT Baseline [29]                                                                             | Sentinel-1 SAR + AMSR2 + ERA5      | SAR Sea Ice Classification Macro $F_1$ : 84.6%   |                       |                             | Pixel Transfer Accuracy: 87.8%                 |                                 |                         | State-of-the-art in [29]          |

TABLE 4  
Quantitative Profiling of Computational Efficiency, Throughput, Hardware Footprint, and Energy vs. Operational NWP Systems.

| Model / Operational System       | Params | Compute Hardware                | Pretraining Cost            | 10-Day Wall-Clock  | Speedup vs. IFS    | Energy / Forecast | Peak Memory Footprint        |
|----------------------------------|--------|---------------------------------|-----------------------------|--------------------|--------------------|-------------------|------------------------------|
| Operational IFS HRES (0.10°)     | N/A    | Atos Sequana HPC (~3,500 CPUs)  | N/A (Numerical PDE)         | ~3,000 s (50 min)  | 1× (Baseline)      | ~180 kWh          | Multi-node cluster (~1.5 TB) |
| Operational IFS ENS (51 members) | N/A    | Atos Sequana HPC (~10,000 CPUs) | N/A (Numerical PDE)         | ~7,200 s (120 min) | 1× (Baseline)      | ~2,200 kWh        | Multi-node cluster (>4.0 TB) |
| Pangu-Weather [6]                | 256M   | 1 × NVIDIA Tesla V100 GPU       | 192 V100 GPUs × 16 days     | 2.0 s              | ~1,500×            | ~0.00015 kWh      | 12.4 GB VRAM                 |
| GraphCast [7]                    | 36.7M  | 1 × Google Cloud TPU v4         | 32 TPU v4 devices × 4 weeks | 58.0 s             | ~50×               | ~0.0050 kWh       | 28.5 GB HBM                  |
| FourCastNet [8]                  | 73.5M  | 1 × NVIDIA A100 GPU             | 64 A100 GPUs × 16 hours     | 0.25 s             | ~12,000×           | ~0.00003 kWh      | 8.2 GB VRAM                  |
| FuXi [32]                        | 150M   | 1 × NVIDIA A100 GPU             | 32 A100 GPUs × 7 days       | 6.0 s              | ~500×              | ~0.0006 kWh       | 18.6 GB VRAM                 |
| AIFS [45]                        | 185M   | 1 × NVIDIA A100 GPU             | 64 A100 GPUs × 14 days      | 20.0 s             | ~150×              | ~0.0018 kWh       | 22.0 GB VRAM                 |
| NeuralGCM (0.70°) [39]           | 14.5M  | 1 × Google Cloud TPU v4         | 4 TPU v4 devices × 5 days   | 30.0 s             | ~100×              | ~0.0028 kWh       | 9.8 GB HBM                   |
| GenCast (50-member ENS) [4]      | 120M   | 8 × Google Cloud TPU v4         | 64 TPU v4 devices × 6 weeks | ~900 s (15 min)    | ~8×                | ~0.85 kWh         | 31.0 GB HBM / worker         |
| CorrDiff (2km regional) [60]     | 110M   | 1 × NVIDIA A100 GPU             | 32 A100 GPUs × 10 days      | 1.5 s              | ~1,000× vs WRF     | ~0.00025 kWh      | 14.2 GB VRAM                 |
| ClimODE (Global continuous) [56] | 0.3M   | 1 × NVIDIA RTX 3090 GPU         | 1 GPU × 24 hours            | 0.4 s              | ~5,000×            | ~0.00004 kWh      | 2.1 GB VRAM                  |
| DGMR (1km radar nowcast) [58]    | 89M    | 1 × Google Cloud TPU v3         | 128 TPU v3 × 3 days         | 0.8 s              | ~250×              | ~0.00012 kWh      | 11.5 GB HBM                  |
| MG-TFNO (Multi-Grid FNO) [34]    | 0.8M   | 1 × NVIDIA A100 GPU             | 1 A100 GPU × 12 hours       | 0.05 s             | ~150× vs Dense FNO | ~0.00001 kWh      | 1.4 GB VRAM                  |
| WaveFNO (Seismic FWI) [35]       | 1.2M   | 1 × NVIDIA RTX 3090 GPU         | 1 GPU × 8 hours             | 2.1 s              | ~1,200× vs FD PDE  | ~0.00018 kWh      | 2.8 GB VRAM                  |
| IceNet (6-Month Arctic) [28]     | 15M    | 1 × NVIDIA Tesla V100 GPU       | 8 V100 GPUs × 4 days        | 1.2 s              | ~2,000× vs SEAS5   | ~0.00009 kWh      | 4.5 GB VRAM                  |